Distributional Learning of Context-Free Languages under Fixed Finite-Semigroup Typing

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Abstract

We study positive-data learning of context-free languages whose nonempty factors are typed by a fixed morphism $s : \Sigma^+ \to \mathsf{S}$ into a finite semigroup. A polynomial-time set-driven operator reconstructs each substitutable target from finite positive witnesses; a separate conservative learner identifies nonempty targets in Gold's sense. The encoded characteristic-data size is polynomial in grammar size and a typed shortest-yield parameter. Ordinary grammar thickness suffices when the typing is determined by a fixed prefix–suffix window and a finite-group value; for linear targets, linear grammar size alone suffices. Conversely, a fixed two-element semilattice typing yields finite targets with linear-size, small-thickness grammars that require exponentially long characteristic samples. The lower bound applies to any set-driven operator admitting characteristic samples for the fixed-typing class. Retyping closure, examples beyond fixed windows, and a Dyck-language obstruction delineate the framework's scope.

Keywords: Grammatical inference; Context-free languages; Distributional learning; Positive-data identification; Finite semigroups

1 Introduction

1.1 Background and motivation

Gold [1] established limitations of positive-data identification. Distributional learning yields positive results for restricted context-free classes, notably Clark–Eyraud substitutability [2] and Yoshinaka's (k, ℓ)-substitutability [3]. For background see de la Higuera [4]; Kanazawa's work on learning categorial grammars offers a complementary approach [5, 6].

Relation to earlier work. The author's 2014 preprint [7] introduced substitutability relative to recognizable equivalence relations, preceding the publication of Yoshinaka's general many-sorted signature framework [8]. For a fixed semigroup morphism $s : \Sigma^+ \to \mathsf{S}$, its nonempty fibers can be represented as disjoint sorts, equipped with typed concatenation and promotion to the whole-string sort. Under this specialization, Yoshinaka's general framework [8, Theorem 1 and Lemma 3] already supplies positive-data identification, polynomial-time hypothesis updates, and characteristic witnesses of polynomial cardinality in a typed target grammar. The particular choice of sorts is not attributed to that work.

The procedures below give an explicit fixed-typing realization, including a direct completeness proof for nullable and unit productions. Their existence, update-time bound, and finite-witness cardinality are not claimed as independent learning-theoretic novelties. Our principal new questions concern the *encoded length* of characteristic data, its polynomial upper and exponential lower bounds, and the algebraic and expressive consequences of finite typing. The locally trivial fixed-window criterion itself is classical.

We study a finite-semigroup-congruence version of this program. Let $s : \Sigma^+ \to \mathsf{S}$ be a homomorphism into a finite semigroup and $\sim_s := \ker s \subseteq \Sigma^+ \times \Sigma^+$. For $x \in \Sigma^*$, write $\mathcal{D}_L(x) := \{(u, v) : uxv \in L\}$. A language L is $\sim_s$-*substitutable* if, for nonempty x, y, $s(x) = s(y)$ and $\mathcal{D}_L(x) \cap \mathcal{D}_L(y) \neq \emptyset$ imply $\mathcal{D}_L(x) = \mathcal{D}_L(y)$. Write $\mathcal{C}_s^{\mathrm{cf}}$ for the context-free $\sim_s$-substitutable languages, and call this the *fixed-s slice*. Thus full distributional equality is required only after a fixed finite algebraic comparison.

For a minimal example, take $\Sigma = \{a\}$ and $L = \{a, aa\}$. Although $(\lambda, \lambda) \in \mathcal{D}_L(a) \cap \mathcal{D}_L(aa)$, the context (a, λ) belongs to $\mathcal{D}_L(a)$ but not to $\mathcal{D}_L(aa)$. Thus L is not Clark–Eyraud substitutable. Under the parity morphism $s(a^n) = n \bmod 2$ into $\mathbb{Z}/2\mathbb{Z}$, however, a and aa have distinct types and $L \in \mathcal{C}_s^{\mathrm{cf}}$.

Finite-index semigroup congruences on Σ^+, equivalently kernels of finite-semigroup homomorphisms, provide finite typings. For each fixed s, write RS_s for the class of all $\sim_s$-substitutable languages, without a context-freeness restriction. For expressive comparison, let $\mathsf{RS} := \bigcup_s \mathsf{RS}_s$, where s ranges over finite-semigroup homomorphisms on Σ^+. Every regular language belongs to this union. Our learning results, however, concern only $\mathcal{C}_s^{\mathrm{cf}}$ for one fixed s; no learner is claimed for RS.

Unlike one fixed prefix–suffix window, s can combine independent finite-state factor features by a product. If the finite typing is allowed to be refined accordingly, the framework is stable under regular filtering and arbitrary inverse homomorphisms, including erasing ones (Proposition 3.2). This is a change-of-typing statement, not closure of one fixed-s slice. An algebraic criterion identifies the typings determined by fixed prefix–suffix windows on nonempty factors (Proposition 3.5); combining windows with finite groups yields a broader characteristic-data bound (Corollary 7.5). Examples below exhibit nonregular linear and nonlinear targets outside every fixed-window class, while counter and Dyck obstructions delimit the reach of fixed finite-semigroup typing.

The choice of s supplies a typing bias on terminal factors [9], not their occurrences. Its connection with regular control sets [10, 11] is confined to the linear setting of Section 8.

1.2 Contributions and main results

We give a finite-witness reconstruction operator Rec_s and a separate conservative Gold learner Learn_s, with polynomial-time reconstruction and updates. These qualitative and polynomial-time guarantees specialize Yoshinaka's general framework [8]. Beyond the classical fixed-window case, our characteristic-data *length* results concern Rec_s; no incremental polynomial-data guarantee is asserted for Learn_s.

The main contributions concern the cost and scope of finite typing:

(1) *Characteristic-data bounds and lower bounds.* For every fixed s, qualitative reconstruction and finite characteristic witnesses work directly for arbitrary CFGs: a general right-hand side contributes one canonical rule witness (Theorem 5.2). For quantitative bounds, conversion to polynomial-size binary–unit–nullable normal form (BUNF) controls the encoded witness length through typed shortest yields (Corollary 6.3). For typings whose positive image lies in the join $\mathbf{LI} \vee \mathbf{Grp}$ of finite locally trivial semigroups and finite groups (Section 7), the size and ordinary thickness of the original CFG suffice (Corollary 7.5); for linear targets, the size of an arbitrary linear CFG alone suffices (Theorem 8.3). Conversely, a fixed two-element semilattice typing gives finite targets with grammar size and ordinary thickness $O(n)$ but characteristic-sample length $\Omega(2^n)$. This lower bound applies to every set-driven operator admitting characteristic samples for both members of a distinguishing pair; the target L_n also has exponential typed thickness under every reduced BUNF presentation (Proposition 6.4; Corollary 6.5).

(2) *Algebraic and expressive scope.* We establish closure under regular filtering and inverse homomorphisms by changing the finite typing, rather than keeping one fixed-s slice. We

record the classical algebraic characterization of fixed-window typings and exhibit non-regular targets beyond all fixed-window classes, together with counter/Dyck obstructions. For linear grammars, typing also induces regular control sets.

2 Preliminaries

Let Σ be a finite alphabet, let Σ^* be the free monoid with identity λ, and put $\Sigma^+ := \Sigma^* \setminus \{\lambda\}$. A pair $(u, v) \in \Sigma^* \times \Sigma^*$ is called a *terminal context*. For $L \subseteq \Sigma^*$ and $x \in \Sigma^*$, the *distribution* of x in L is $\mathcal{D}_L(x) := \{(u, v) \in \Sigma^* \times \Sigma^* : uxv \in L\}$. The *syntactic congruence* of L is defined by $x \equiv_L y$ iff $\mathcal{D}_L(x) = \mathcal{D}_L(y)$, for $x, y \in \Sigma^*$ [12, Chapter III, Proposition 10.2]. As usual, $\equiv_L$ is a monoid congruence: if $x \equiv_L x'$ and $y \equiv_L y'$, then $xy \equiv_L x'y'$.

For a word $w \in \Sigma^*$ and a letter $a \in \Sigma$, let $|w|_a$ denote the number of occurrences of a in w.

A context-free grammar (CFG) is a tuple $G = (V, \Sigma, P, S)$ with finite nonterminal set V, start symbol $S \in V$, and productions $A \to \alpha$ with $A \in V$ and $\alpha \in (V \cup \Sigma)^*$. A *sentential form* is any word in $(V \cup \Sigma)^*$. We write $\alpha \Rightarrow_G \beta$ for one derivation step in G, and $\Rightarrow_G^*$ for its reflexive-transitive closure. Thus $L(G) := \{w \in \Sigma^* : S \Rightarrow_G^* w\}$. For $A \in V$, put $L_G(A) := \{w \in \Sigma^* : A \Rightarrow_G^* w\}$. For a nonterminal $X \in V$, a terminal context (u, v) is a *terminal reaching context of X in G* if $S \Rightarrow_G^* uXv$. A nonterminal is *reachable* if it occurs in a sentential form reachable from S, *productive* if it derives a terminal word, *useless* if it is not both, and *nullable* if it derives λ; G is *reduced* if it has no useless nonterminal. A *unit production* (*unit rule*) is $A \to B$ with $A, B \in V$, and the *unit closure* is the reflexive-transitive closure of the induced relation on V. For $L(G) \neq \emptyset$, the *trim* (*reduced part*) of G deletes the nonproductive nonterminals and then the unreachable ones, each with its incident productions. A CFG is *linear* if every right-hand side contains at most one nonterminal. By *binarization* we mean replacement of longer right-hand sides by equivalent chains of binary productions using fresh nonterminals. We write CFL for the class of context-free languages. For any finite representation $\mathcal{R}$, $|\mathcal{R}|$ denotes the length of a fixed reasonable encoding; for a grammar this is polynomially equivalent to the usual symbol count of its productions. Following the thickness parameter used by Wakatsuki–Tomita [13] and adopted by Yoshinaka [3] (see also Eyraud–Heinz–Yoshinaka [14, Definition 2.11]), for a reduced CFG $G = (V, \Sigma, P, S)$ we use the shortest-yield *thickness*

$$\tau_G := \max_{A \in V} \min\{\, |w| : A \Rightarrow_G^* w,\ w \in \Sigma^* \,\}.$$

2.1 Learning from positive data

We use Gold's explanatory model [1]. Let H_0 denote a fixed CFG with one start symbol and no productions, so $L(H_0) = \emptyset$. We use texts only for nonempty targets and represent the empty language separately by H_0; equivalently one may adjoin the usual pause symbol. For a nonempty language L, a *text* is an infinite sequence $\mathcal{T} = (w_1, w_2, \ldots)$ over L in which every word of L occurs. Write $\mathcal{T}[n] = (w_1, \ldots, w_n)$ and $K_n := \{w_1, \ldots, w_n\}$. A learner Learn *identifies a nonempty L in the limit* if for every text $\mathcal{T}$ for L there are n_0 and a grammar H such that $\mathsf{Learn}(\mathcal{T}[n]) = H$ for every $n \geq n_0$ and $L(H) = L$. Thus the hypothesis representation itself eventually stabilizes. A sequential learner is *conservative* if it changes its current hypothesis only when the newly presented positive datum is not generated by that hypothesis.

We distinguish this sequential learner from a *set-driven reconstruction operator* Rec: this is a map from finite sets of words to grammars; in particular, its output is independent of presentation order and multiplicity. We call Rec *sample-consistent* if $K \subseteq L(\mathsf{Rec}(K))$ for every finite sample K. A finite set $\mathcal{F} \subseteq L$ is a *characteristic sample of L for* Rec if $\mathcal{F} \subseteq K \subseteq L$ implies $L(\mathsf{Rec}(K)) = L$ for every finite K; this is the positive-data, set-driven specialization of the characteristic samples of de la Higuera [15] (cf. [4, Section 7.3.3, Definition 7.3.3], formulated for sequential presentations). We measure sample size by $\|K\| := \sum_{w \in K} (|w| + 1)$, so that λ counts and $|K| \leq \|K\|$, and call $\|\mathcal{F}\|$ the *characteristic data* of $\mathcal{F}$. Polynomial update time

and polynomial characteristic data are distinct criteria [4, Sections 7.3.2–7.3.3]. In the terms of Eyraud–Heinz–Yoshinaka [14, Definitions 2.5 and 2.8], our polynomial characteristic-data guarantees are *set-based* (correct reconstruction by Rec_s from every finite sample containing a characteristic sample); the *incremental* criterion, a correct and stable hypothesis of Learn_s as soon as such a sample has appeared, is not claimed. Gold stabilization of Learn_s is proved in Corollary 5.4.

2.2 Fixed finite-semigroup typing

Fix a finite semigroup S and a *surjective* semigroup homomorphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$, represented by the multiplication table of S and the letter images $s(a)$ for $a \in \Sigma$. Thus $s(w)$ is computed by a left-to-right product in $O(|w|)$ time for nonempty w. Surjectivity is without loss of generality: an arbitrary finite semigroup typing may be replaced by its positive image. Whenever a later construction produces a non-surjective semigroup map (notably a product or inverse-image typing), its type codomain is implicitly restricted to the positive image before taking the fresh-identity completion. We work over a nonempty alphabet unless stated otherwise; for $\Sigma = \emptyset$, the only possible target languages are $\emptyset$ and $\{\lambda\}$, which the unchanged learner handles directly.

For a finite semigroup T, an element $e \in \mathsf{T}$ is *idempotent* if $e^2 = e$; write $E(\mathsf{T})$ for its idempotents. Call T *locally trivial* when $ete = e$ for all $e \in E(\mathsf{T})$ and $t \in \mathsf{T}$. Every nonempty finite semigroup has an idempotent. Consequently local triviality is preserved by surjective finite-semigroup morphisms: if $\pi : \mathsf{T} \to \mathsf{S}$ is surjective, every idempotent of S lifts to one in its preimage subsemigroup.

Definition 2.1 (fixed-s substitutability)**.** *A language $L \subseteq \Sigma^*$ is* $\sim_s$-substitutable *if for all $x, y \in \Sigma^+$, equality $s(x) = s(y)$ together with $\mathcal{D}_L(x) \cap \mathcal{D}_L(y) \neq \varnothing$ implies $\mathcal{D}_L(x) = \mathcal{D}_L(y)$.*

Lemma 2.2 (canonical fresh-identity completion)**.** *Let $\mathsf{S}^\sharp = \mathsf{S} \sqcup \{\mathbf{1}_\sharp\}$ be the monoid obtained by adjoining a* fresh *identity $\mathbf{1}_\sharp$ to the finite semigroup S, even if S already has an identity. There is a unique monoid extension $\widehat{s} : \Sigma^* \twoheadrightarrow \mathsf{S}^\sharp$ of s, given by*

$$\widehat{s}(w) = \begin{cases} \mathbf{1}_\sharp, & w = \lambda, \\ s(w), & w \in \Sigma^+. \end{cases}$$

It is surjective, and $\mathsf{S}^\sharp \cong \Sigma^* / \ker \widehat{s}$, with

$$\widehat{s}(x) = \widehat{s}(y) \quad \Longleftrightarrow \quad (x = y = \lambda) \text{ or } (x, y \in \Sigma^+ \text{ and } s(x) = s(y)).$$

Thus Definition 2.1 is equivalent to the all-fragment condition

$$\forall x, y \in \Sigma^*, \quad \big(\widehat{s}(x) = \widehat{s}(y) \wedge \mathcal{D}_L(x) \cap \mathcal{D}_L(y) \neq \emptyset\big) \Longrightarrow \mathcal{D}_L(x) = \mathcal{D}_L(y).$$

Conversely, every finite-monoid typing $h : \Sigma^* \to M$ induces the finite-semigroup typing $s = h|_{\Sigma^+} : \Sigma^+ \to h(\Sigma^+)$. If substitutability relative to h compares nonempty *factors, as in Definition 2.1, it defines exactly the same class and nonempty-factor reconstruction. No equivalence is asserted for the different condition that also compares λ with nonempty factors of type $h(\lambda)$.*

Proof. Adjoining the identity extends multiplication and makes $\widehat{s}$ a monoid morphism. Every element of S is attained by a positive word, while λ attains $\mathbf{1}_\sharp$, so the map is surjective. The displayed kernel description follows because $\mathbf{1}_\sharp \notin \mathsf{S}$. Equal extended types compare either λ with itself or nonempty words of equal s-type, proving equivalence with Definition 2.1. For the converse, restricting h preserves all nonempty equalities, which alone define substitutability and the reconstruction rules. □

Throughout, subscripts of substitutability classes, reconstruction operators, and typed-thickness parameters denote semigroup morphisms on *nonempty* words. Write s_0 for the unique one-element semigroup typing, on whichever alphabet is in use. When a monoid morphism $h : \Sigma^* \to M$ is convenient for recognizing a regular language or evaluating empty contexts, its restriction $s = h|_{\Sigma^+}$ is the learning bias. The extension $\widehat{s}$ handles nullable yields (Lemma 2.2); algebraic conditions are tested on the positive image, not on its completion.

For $x \in \Sigma^*$ we call $\widehat{s}(x)$ its *yield type*; for nonempty x it equals $s(x)$. As above, $\mathcal{C}_s^{\mathrm{cf}}$ denotes the context-free $\sim_s$-substitutable languages, and we define $\mathcal{C}_s^{\mathrm{lin}} := \{L \in \mathcal{C}_s^{\mathrm{cf}} : L \text{ is generated by some linear CFG}\}$.

3 Fixed Recognizable Substitutability

For fixed s, put $\mathsf{RS}_s := \{L \subseteq \Sigma^* : L \text{ is } \sim_s\text{-substitutable}\}$. Then the fixed-$s$ slice is $\mathcal{C}_s^{\mathrm{cf}} = \mathsf{RS}_s \cap \mathrm{CFL}$, while $\mathcal{C}_s^{\mathrm{lin}}$ consists of its linearly generated members. For expressive comparisons only, let $\mathsf{RS} := \bigcup_s \mathsf{RS}_s$, where s ranges over finite-semigroup homomorphisms on Σ^+. We write $\mathsf{RS}_s(\Gamma)$ and $\mathsf{RS}(\Gamma)$ when the alphabet matters; all inclusions below are alphabetwise. The learning results always concern one fixed s.

Proposition 3.1. *A language $L \subseteq \Sigma^*$ is regular iff $L = \widehat{s}^{-1}(\mathsf{Acc})$ for some finite-semigroup typing $s : \Sigma^+ \to \mathsf{S}$ and some $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$. Every language recognized in this way belongs to RS_s, so all regular languages belong to RS.*

Proof. For regular L, restrict its syntactic-monoid morphism [12, Chapter III, Proposition 10.2] to Σ^+, replace the codomain by its positive image S, and extend by a fresh identity. The positive types still determine membership for nonempty words; include $\mathbf{1}_\sharp$ among the accepting types exactly if $\lambda \in L$. Conversely, $\widehat{s}$ recognizes a regular language since $\mathsf{S}^\sharp$ is finite. For nonempty x, y with $s(x) = s(y)$, $\widehat{s}(uxv) = \widehat{s}(uyv)$ for all contexts $u, v \in \Sigma^*$, hence $\mathcal{D}_L(x) = \mathcal{D}_L(y)$. □

By Proposition 3.1, RS and $\mathsf{RS} \cap \mathrm{CFL}$ contain every finite language and Σ^*; Gold's theorem makes both unions nonlearnable from positive data on nonempty alphabets. Yoshinaka [3, Section 4] notes the analogous issue for the union of fixed-window classes. We learn only fixed-s slices.

The fresh-identity completion reduces refinement and product typing to ordinary monoid-kernel operations. For finite-semigroup typings s, s', t of Σ^+, where each codomain is reduced to its positive image,

$$\ker \widehat{s'} \subseteq \ker \widehat{s} \iff \ker s' \subseteq \ker s, \qquad \ker \widehat{s \times t} = \ker \widehat{s} \cap \ker \widehat{t}.$$

Here $\widehat{s \times t}$ completes the *image* $(s \times t)(\Sigma^+)$ with a fresh identity. In fact its image monoid is isomorphic to $\mathrm{im}(\widehat{s} \times \widehat{t})$, a submonoid of $\mathsf{S}^\sharp \times \mathsf{T}^\sharp$. Consequently $\mathsf{RS}_s \subseteq \mathsf{RS}_{s'}$ under the first inclusion, and $\mathsf{RS}_s \cup \mathsf{RS}_t \subseteq \mathsf{RS}_{s \times t}$. A refinement may increase the typed grammar size used in quantitative bounds.

Yoshinaka [3, Proposition 1] showed that fixed-window substitutable languages are not closed under intersection with regular languages or inverse images under homomorphisms that may erase. In contrast, they are closed under inverse images of nonerasing homomorphisms: for such a morphism φ, equality of fixed k-prefixes and ℓ-suffixes is preserved by φ. Indeed, $|\varphi(p)| \geq k$ for $p \in \Gamma^k$, so the first k letters of $\varphi(prq)$ depend only on p, and the suffix case is symmetric; short factors of equal window type are identical. Thus the fixed-window substitutability implication pulls back along φ. The following proposition shows that regular filtering and erasing inverse images can be accommodated by changing the finite typing.

Proposition 3.2 (closure under finite information)**.** *Let $s : \Sigma^+ \to \mathsf{S}$ and $t : \Sigma^+ \to \mathsf{T}$ be finite-semigroup typings.*

(i) *If* $L \in \mathsf{RS}_s$ *and* $J \in \mathsf{RS}_t$*, then* $L \cap J \in \mathsf{RS}_{s\times t}$.

(ii) *If* $R_{\mathrm{reg}} = \widehat{t}^{-1}(\mathsf{Acc}_t)$ *is regular and* $L \in \mathcal{C}_s^{\mathrm{cf}}$*, then* $L \cap R_{\mathrm{reg}} \in \mathcal{C}_{s\times t}^{\mathrm{cf}}$.

(iii) *Let* $\varphi : \Gamma^* \to \Sigma^*$ *be any homomorphism, possibly erasing, and put* $r = (\widehat{s}\circ\varphi)|_{\Gamma^+} : \Gamma^+ \to \mathsf{S}^\sharp$. *If* $L \in \mathsf{RS}_s$*, then* $\varphi^{-1}(L) \in \mathsf{RS}_r$*; if moreover* $L \in \mathcal{C}_s^{\mathrm{cf}}$*, then* $\varphi^{-1}(L) \in \mathcal{C}_r^{\mathrm{cf}}$.[1]

Proof. For (i), nonempty factors of equal $(s \times t)$-type with a common context in $L \cap J$ have equal distributions in each language, and hence in their intersection. Item (ii) follows from (i) and closure of CFLs under regular intersection. For (iii), r is a semigroup morphism, since $\widehat{s}\circ\varphi$ is a monoid morphism. If two nonempty factors of equal r-type share a context in $\varphi^{-1}(L)$, their images are either both empty or both nonempty and of equal s-type. In the first case their inverse-image distributions coincide. In the second case, substitutability of L and the shared image context give the same conclusion after pulling back contexts. The context-free assertion uses closure of CFLs under inverse homomorphism. □

Typing adequacy. By Lemma 2.2, $L \in \mathsf{RS}_s$ exactly when $\ker \widehat{s}$ distinguishes every pair of factors with intersecting but unequal distributions. Thus typing adequacy is a finite-index congruence separation condition, preserved under refinement and products at a possible quantitative cost (Proposition 6.4). The stronger condition $\ker \widehat{s} \subseteq \equiv_L$ is sufficient but forces L to be regular.

3.1 Classical special cases

Clark–Eyraud's original Definition 5 includes all fragments [2]; we follow Clark [16] in excluding λ. This distinction matters even for $L = a^+$: (a, λ) belongs to both $\mathcal{D}_L(\lambda)$ and $\mathcal{D}_L(a)$, whereas (λ, λ) belongs to $\mathcal{D}_L(a)$ but not to $\mathcal{D}_L(\lambda)$. Thus comparing λ with nonempty factors *without type separation* would exclude a^+; $\widehat{s}$ makes those factors incomparable in type. Call L *Clark–Eyraud substitutable* when $\mathcal{D}_L(x) \cap \mathcal{D}_L(y) \neq \emptyset$ implies $\mathcal{D}_L(x) = \mathcal{D}_L(y)$ for all $x, y \in \Sigma^+$. This is exactly the trivial-semigroup case of Definition 2.1, and hence also Yoshinaka's $(0, 0)$ case.

For $k, \ell \geq 0$, let $\mathrm{pref}_k(w)$ and $\mathrm{suf}_\ell(w)$ denote the prefix and suffix of lengths $\min\{k, |w|\}$ and $\min\{\ell, |w|\}$, respectively; in particular, $\mathrm{pref}_0(w) = \mathrm{suf}_0(w) = \lambda$. Renaming Yoshinaka's variables so that u_i, v_i denote outer contexts, L is (k, ℓ)-*substitutable* if, for all $p \in \Sigma^k$, $q \in \Sigma^\ell$, and $u_1, u_2, v_1, v_2, r_1, r_2 \in \Sigma^*$ with $pr_1q, pr_2q \neq \lambda$,

$$u_1pr_1qv_1 \in L, \quad u_1pr_2qv_1 \in L, \quad u_2pr_1qv_2 \in L \\ \implies \quad u_2pr_2qv_2 \in L.$$

Proposition 3.3 (finite-semigroup realization of fixed windows)**.** *For every* $k, \ell \geq 0$ *there is a surjective finite-semigroup morphism* $s_{k,\ell} : \Sigma^+ \to W_{k,\ell}$ *such that* $L \subseteq \Sigma^*$ *is* (k, ℓ)*-substitutable if and only if it is* $\sim_{s_{k,\ell}}$*-substitutable.*

Proof. Put $m_0 := \max\{1, k + \ell\}$. For $w \in \Sigma^+$ define

$$s_{k,\ell}(w) := \begin{cases} (\mathrm{short}, w), & 0 < |w| < m_0, \\ (\mathrm{long}, \mathrm{pref}_k(w), \mathrm{suf}_\ell(w)), & |w| \geq m_0. \end{cases}$$

The displayed equivalence on Σ^+ is a two-sided congruence: equal short types represent the same word; equal long types retain their length-k prefix and length-ℓ suffix after any common left and right concatenation. Its finite quotient $W_{k,\ell}$ thus inherits an associative product, and $s_{k,\ell}$ is a surjective semigroup morphism.

For nonempty x, y, equal types mean either $x = y$ or $x = pr_1q$, $y = pr_2q$ for common $p \in \Sigma^k$, $q \in \Sigma^\ell$ and $r_1, r_2 \in \Sigma^*$. In the latter case a common context and an arbitrary context of one factor instantiate Yoshinaka's four-word condition; symmetry gives equal distributions. Conversely, the

[1] The fresh identity in $\mathsf{S}^\sharp$ replaces a separate erasure flag. For trivial s, $L = a^+$, $\varphi(a) = a$ and $\varphi(b) = \lambda$, the two nonempty inverse-image factors a, b share a context but have different distributions.

nonempty factors of any such premise have equal $s_{k,\ell}$-types, so relative substitutability supplies the fourth word. □

Yoshinaka [8, Example 4] already represented (k,ℓ)-substitutability by the sorted signature $\Sigma_{k,\ell}$. Proposition 3.3 makes the inducing finite-semigroup quotient explicit; no priority for the window-signature encoding is claimed.

Fixed-window counterexample criterion. If x, y have length at least $\max\{1, k+\ell\}$, the same length-k prefix and length-ℓ suffix, a common context, and unequal distributions, then L is not (k,ℓ)-substitutable: the first three conditions give equal $s_{k,\ell}$-types and a shared context, while the last contradicts $\sim_{s_{k,\ell}}$-substitutability.

Thus the fixed (k,ℓ) class is exactly $\mathsf{RS}_{s_{k,\ell}}$. Section 7 gives its polynomial thick-data bound. For $k = \ell = 0$, $s_{0,0}$ maps all nonempty words to one type, so it coincides with s_0 on Σ^+. For an alphabet Γ, set

$$\mathsf{KL}(\Gamma) := \bigcup_{k,\ell \geq 0} \{L \subseteq \Gamma^* : L \text{ is } (k,\ell)\text{-substitutable}\},$$

and abbreviate $\mathsf{KL} := \mathsf{KL}(\Sigma)$.

3.2 Main theorem

Theorem 3.4 (fixed-s learning theorem)**.** *Fix an efficiently computable finite-semigroup homomorphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$. There are a set-driven reconstruction operator Rec_s and a conservative sequential learner Learn_s such that:*

(i) *Rec_s is sample-consistent and $\mathsf{Rec}_s(K)$ is constructible in time polynomial in $\|K\|$;*

(ii) *every $L \in \mathcal{C}_s^{\mathrm{cf}}$ has a finite characteristic sample for Rec_s;*

(iii) *Learn_s identifies every nonempty $L \in \mathcal{C}_s^{\mathrm{cf}}$ in Gold's sense, each update taking time polynomial in the data seen so far;*

(iv) *if the positive image semigroup $s(\Sigma^+)$ is a homomorphic image of a subsemigroup of the direct product of a finite locally trivial semigroup and a finite group, then for every nonempty target the sample in (ii) can be chosen polynomial in $|G_*|$ and $\tau_{G_*}+1$ for any reduced CFG G_* representing that target; this includes every fixed-window and every finite-group typing;*

(v) *for arbitrary fixed s and $L \in \mathcal{C}_s^{\mathrm{lin}}$, the sample in (ii) can be chosen polynomial in $|G_*|$ for any linear CFG G_* representing L.*

The polynomials in (iv)–(v) may depend on the fixed s and Σ, but not on the chosen target grammar G_*. The quantitative mechanism behind items (iv)–(v), together with its limitation for arbitrary fixed typings, is developed in Sections 6–8.

This is consistent with Gold's theorem: on a nonempty alphabet choose distinct $x, y \in \Sigma^+$ with $s(x) = s(y)$, and choose $z \in \Sigma^*$ with $|z| > \max\{|x|, |y|\}$. The finite language $\{x, y, zx\}$ is not $\sim_s$-substitutable: x, y share the empty context but only x admits (z, λ).

Algebraic position of fixed windows. The locally trivial-semigroup identities underlying the following criterion and the corresponding prefix–suffix characterization of recognizable languages are classical [17, Chapter 2, Proposition 3.5 and Theorem 3.6]; see also the generalized definite language characterization [18, Proposition 5.18]. We state the positive-image/kernel formulation explicitly for arbitrary finite-semigroup typings and prove it below. Its learning-specific application is Corollary 7.5. These algebraic conditions must be tested on $\mathsf{S} = s(\Sigma^+)$, *not* on $\mathsf{S}^\sharp$. Indeed, for nonempty S, the new identity $\mathbf{1}_\sharp$ and any $t \in \mathsf{S}$ satisfy $\mathbf{1}_\sharp t \mathbf{1}_\sharp = t \neq \mathbf{1}_\sharp$; thus $\mathsf{S}^\sharp$ itself is not locally trivial.

Proposition 3.5 (locally trivial typings and fixed windows)**.** *Let $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ be a finite-semigroup typing on a nonempty alphabet and set $n := |\mathsf{S}| + 1$. The following are equivalent:*

(i) S *is locally trivial;*

(ii) $\ker s_{k,\ell} \subseteq \ker s$ *for some* $k, \ell \geq 0$.

In (ii), $k = \ell = n$ *suffices. Every window semigroup* $W_{k,\ell} = s_{k,\ell}(\Sigma^+)$ *is locally trivial. Consequently,*

$$\mathsf{KL}(\Sigma) = \bigcup_{\substack{s:\Sigma^+\to\mathsf{S} \\ \mathsf{S}\ \textit{finite locally trivial}}} \mathsf{RS}_s(\Sigma),$$

also after intersecting both sides with CFL.

Proof. A short-word class in $W_{k,\ell}$ cannot be idempotent: squaring a positive word either changes its short representative or reaches the long part. If $e = s_{k,\ell}(x)$ is a long type, then xux retains the prefix and suffix of x for every $u \in \Sigma^+$. Thus $ete = e$ for all $t \in W_{k,\ell}$.

For a locally trivial S, the classical identities $\mathsf{S}^n = \mathsf{S}E(\mathsf{S})\mathsf{S}$ and $etf = ef$ for idempotents e, f and all $t \in \mathsf{S}$ hold [17, Chapter 1, Proposition 1.12; Chapter 2, Proposition 3.5]. For $p, q \in \Sigma^n$, write $s(p) = aeb$, $s(q) = cfd$ with e, f idempotent. For every $r \in \Sigma^*$, including λ, $b\widehat{s}(r)c \in \mathsf{S}$, so $s(prq) = a\,e\,(b\widehat{s}(r)c)\,f\,d = aefd = s(pq)$. Hence length-$n$ windows determine the type of every word of length at least $2n$; shorter words are separated by $s_{n,n}$. Conversely, (ii) induces a surjective semigroup morphism $W_{k,\ell} \to \mathsf{S}$, preserving local triviality. The union identities follow by kernel refinement. □

4 Reconstruction Operator and Sequential Learner

4.1 Finite-sample reconstruction

For a finite sample $K \subseteq \Sigma^*$, let

$$\widehat{V}(K) := \{[\![x]\!] : x \in \Sigma^+,\ \exists u, v \in \Sigma^* \text{ with } uxv \in K\}.$$

Thus there is one hypothesis nonterminal for each distinct nonempty factor observed in the sample; the double brackets are part of the nonterminal name and do not denote a quotient class or literal terminal symbols. For observed nonempty factors x, y, write

$$x \approx_{K,s} y \quad\Longleftrightarrow\quad s(x) = s(y) \ \text{ and } \ \mathcal{D}_K(x) \cap \mathcal{D}_K(y) \neq \emptyset.$$

The grammar $\mathsf{Rec}_s(K)$ has nonterminal set $\widehat{V}(K) \cup \{\widehat{S}\}$, start symbol $\widehat{S}$, and the following *set* of productions whenever every displayed bracketed nonterminal belongs to $\widehat{V}(K)$:

$$\begin{aligned}
&\text{(B)} && [\![xy]\!] \to [\![x]\!][\![y]\!], && x, y \in \Sigma^+,\\
&\text{(U)} && [\![x]\!] \to [\![y]\!], && x \approx_{K,s} y,\\
&\text{(L)} && [\![a]\!] \to a, && a \in \Sigma,\\
&\text{(S)} && \widehat{S} \to [\![w]\!], && w \in K \cap \Sigma^+.
\end{aligned}$$

If $\lambda \in K$, Rule (S) also contains $\widehat{S} \to \lambda$.

For finite samples $K \subseteq K'$, every nonterminal and production of $\mathsf{Rec}_s(K)$ also occurs in $\mathsf{Rec}_s(K')$. Consequently, $L(\mathsf{Rec}_s(K)) \subseteq L(\mathsf{Rec}_s(K'))$.

The construction follows the substring-indexed grammar scheme of Yoshinaka [3, Section 4, Algorithm 1], also recalled by Clark [16, Section 4, Algorithm 1]. For the semigroup typing $s = s_{k,\ell}$, the nontrivial instances of Rule (U) coincide with Yoshinaka's unary rules: distinct observed nonempty factors are connected when they share a sample context and have equal $s_{k,\ell}$-type; equivalently, both factors have length at least $\max\{1, k + \ell\}$ and have the same length-k prefix and length-ℓ suffix. The constructions agree up to redundant identity rules, the representation of initial productions by a separate start symbol, and the treatment of λ. For general fixed s, the prefix–suffix test is replaced by equality of s-types.

Lemma 4.1 (sample consistency)**.** *For every finite sample K, $K \subseteq L(\mathsf{Rec}_s(K))$.*

Proof. If $w = a_1 \cdots a_m \in K \cap \Sigma^+$, Rule (S) enters $[\![w]\!]$. Repeated use of Rule (B) along any binary factorization of w reaches the one-letter factors, all of which are observed because they occur inside the same sample word. Rule (L) then emits $a_1 \cdots a_m$. If $\lambda \in K$, the start rule emits it directly. □

Theorem 4.2 (soundness)**.** *If $K \subseteq L$ and L is $\sim_s$-substitutable, then $L(\mathsf{Rec}_s(K)) \subseteq L$.*

Proof. Induct on the height of a derivation tree rooted at $[\![x]\!]$ to prove

$$[\![x]\!] \Rightarrow^*_{\mathsf{Rec}_s(K)} w \quad \Longrightarrow \quad x \equiv_L w \ \text{ and } \ s(w) = s(x). \tag{1}$$

Every factor nonterminal is indexed by a nonempty observed word, so the fragments compared by Rule (U) satisfy the nonempty requirement of Definition 2.1. Moreover, Rules (B)–(L) cannot derive λ from a factor nonterminal. Rule (L) is immediate. For Rule (U), suppose $[\![x]\!] \to [\![y]\!]$. By definition there is a sample context (u, v) with $uxv, uyv \in K \subseteq L$ and $s(x) = s(y)$. Fixed-s substitutability gives $x \equiv_L y$; the induction hypothesis gives $y \equiv_L w$ and $s(w) = s(y)$, proving (1).

For Rule (B), write the derived word as $w_1 w_2$ from $[\![xy]\!] \to [\![x]\!][\![y]\!]$. The induction hypotheses give $x \equiv_L w_1$, $y \equiv_L w_2$, $s(w_1) = s(x)$, and $s(w_2) = s(y)$. Because $\equiv_L$ is a monoid congruence, $xy \equiv_L w_1 w_2$, and the homomorphism law gives $s(w_1 w_2) = s(xy)$.

Finally, if a start rule $\widehat{S} \to [\![x]\!]$ is used, then $x \in K \subseteq L$, so $(\lambda, \lambda) \in \mathcal{D}_L(x)$. The invariant gives $\mathcal{D}_L(w) = \mathcal{D}_L(x)$ and hence $w \in L$. The direct λ rule is present only when $\lambda \in K \subseteq L$. □

4.2 The sequential learner

Recomputing Rec_s after every datum need not stabilize syntactically: for $L_c = \{a^n c b^n : n \geq 0\}$, longer examples introduce new factor nonterminals such as $[\![a^n c b^n]\!]$. We therefore use the conservative wrapper of Clark–Eyraud's Substitution Graph Learner (SGL, Algorithm 2) and Yoshinaka's (k, ℓ)-Substitutable Grammar Learner ((k, ℓ)-SGL, Algorithm 1) [2, 3]; see also Eyraud–Heinz–Yoshinaka [14, Appendix] for the distinction between set-driven reconstruction and incremental learning. Starting from the fixed empty grammar H_0, after reading w_n put

$$H_n := \begin{cases} H_{n-1}, & w_n \in L(H_{n-1}), \\ \mathsf{Rec}_s(K_n), & w_n \notin L(H_{n-1}), \end{cases} \qquad \mathsf{Learn}_s(\mathcal{T}[n]) := H_n.$$

Every rebuild uses the whole accumulated sample K_n. Although Rec_s is set-driven, Learn_s can stabilize on different grammars for different presentation orders. Indeed, let $\mathcal{T}$ be a text for an infinite target and choose a word $w \in L$ longer than every sample word at its last rebuild. The limiting grammar for $\mathcal{T}$ has no symbol $[\![w]\!]$, whereas the text obtained by prepending w rebuilds on a sample containing w and therefore retains $[\![w]\!]$ in its limiting grammar. Both hypotheses generate L; only eventual syntactic stabilization is asserted (Corollary 5.4). See also Clark [16, Section 4, Example 1].

5 Finite Witnesses, Exact Reconstruction, and Complexity

5.1 Target refinement and finite witnesses

For qualitative completeness, type an arbitrary target CFG over $\mathsf{S}^\sharp$. The hypothesis grammar $\mathsf{Rec}_s(K)$ remains unchanged, using nonempty observed factors and a direct start rule for λ.

Let $G = (V, \Sigma, P, S_0)$ be a CFG for a nonempty L. Write a rule as $\rho : A \to u_0 B_1 u_1 \cdots B_r u_r$ with $u_i \in \Sigma^*$, $B_i \in V$, and $r \geq 0$. For each $(\mu_1, \ldots, \mu_r) \in (\mathsf{S}^\sharp)^r$, set $\mu = \widehat{s}(u_0) \prod_{i=1}^r (\mu_i \widehat{s}(u_i))$ and include

$$A_\mu \to u_0 (B_1)_{\mu_1} u_1 \cdots (B_r)_{\mu_r} u_r.$$

This also covers terminal-only, empty, and unit rules. Add a fresh start symbol $\widetilde{S}$ with $\widetilde{S} \to (S_0)_\mu$ for each $\mu \in \mathsf{S}^\sharp$, and trim to the reduced grammar $\widetilde{G}$. This refinement is finite but can have exponential size; it is not part of the learning algorithm.

Every retained typed symbol is reachable and productive. Its yields have the indicated type; in particular, $A_{\mathbf{1}_\sharp}$ yields only λ, while A_μ with $\mu \neq \mathbf{1}_\sharp$ yields no empty word. For each retained non-start symbol X, choose the shortlex-minimal yield $\omega(X)$, possibly λ. Typing ensures that $\omega(X)$ and every canonical right-hand-side yield of X share their type. Surrounding contexts carry no type annotations.

Proposition 5.1 (typed lifting and yield invariant)**.** *The typed refinement of any CFG G satisfies $L(\widetilde{G}) = L(G)$. For every retained A_μ, including typed copies of S_0,*

$$L_{\widetilde{G}}(A_\mu) = L_G(A) \cap \widehat{s}^{-1}(\{\mu\}).$$

In particular, $A_\mu \Rightarrow^*_{\widetilde{G}} w$ entails $\widehat{s}(w) = \mu$.

Proof. Erasing types gives a G-derivation; conversely, annotating each node by its terminal-yield type gives a typed derivation, including nullable children. Successful typed derivations survive trimming. For a retained A_μ, graft any annotated derivation of $w \in L_G(A)$ of type μ at a reachable occurrence of A_μ. This establishes the fiber and start-language equalities. □

For each retained non-start symbol X, choose a *canonical reaching context* $\chi(X) = (u_X, v_X)$ minimizing $|u_X| + |v_X|$, with shortlex tie breaks on u_X and then v_X. It exists because $\widetilde{G}$ was trimmed. For each typed rule $\rho : X \to u_0 Y_1 u_1 \cdots Y_r u_r$ ($X \neq \widetilde{S}$), let its *canonical right-hand-side yield* be $t_\rho = u_0 \omega(Y_1) u_1 \cdots \omega(Y_r) u_r$. Both t_ρ and $\omega(X)$ belong to $L_{\widetilde{G}}(X)$ and have the same type. Define the finite set of *canonical positive witnesses*

$$\begin{aligned} \mathsf{Wit}(\widetilde{G}) := {} & \{u_X \omega(X) v_X : X \in \widetilde{V} \setminus \{\widetilde{S}\}\} \\ & \cup \{u_X t_\rho v_X : \rho = (X \to \alpha) \in \widetilde{P},\ X \neq \widetilde{S}\}. \end{aligned} \tag{2}$$

Every witness belongs to L. For an empty-type rule, $t_\rho = \omega(X) = \lambda$, so its witness is redundant. If $\lambda \in L$, the start copy $(S_0)_{\mathbf{1}_\sharp}$ has context (λ, λ) and yield λ; hence $\lambda \in \mathsf{Wit}(\widetilde{G})$.

5.2 Exact reconstruction and Gold convergence

Theorem 5.2 (completeness from the finite witnesses)**.** *Let G be any CFG generating a nonempty language L, and $\widetilde{G}$ its $\mathsf{S}^\sharp$-typed refinement. If $\mathsf{Wit}(\widetilde{G}) \subseteq K$, then $L \subseteq L(\mathsf{Rec}_s(K))$.*

Neither substitutability nor $K \subseteq L$ is required.

Proof. For retained $X = A_\mu$ of nonempty type, put $\widehat{X} := [\![\omega(X)]\!]$, an observed nonempty factor by the first witness family. Induct on the height of $X \Rightarrow^*_{\widetilde{G}} w$ to show $\widehat{X} \Rightarrow^*_{\mathsf{Rec}_s(K)} w$.

For the root rule $\rho : X \to u_0 Y_1 u_1 \cdots Y_r u_r$, $\omega(X)$ and t_ρ are nonempty of the same s-type and occur in context (u_X, v_X) in K. Rule (U) gives $\widehat{X} \to [\![t_\rho]\!]$. List the nonempty components of t_ρ in order as $z_1, \ldots, z_h$, where $h \geq 1$: they are nonempty terminal blocks and the $\omega(Y_i)$ of positive-type children. Empty-type children yield only λ and are omitted. Each component and each consecutive concatenation is an observed nonempty factor of $u_X t_\rho v_X \in K$. Repeated Rule (B) thus splits $[\![t_\rho]\!]$ into $[\![z_1]\!], \ldots, [\![z_h]\!]$. Terminal blocks are emitted by Rules (B) and (L); positive-type children use the induction hypothesis. This produces w, including for unit, nullable and mixed productions.

Every positive target word has a typed start copy $(S_0)_\mu$ whose context is (λ, λ). Its canonical yield belongs to K, allowing start Rule (S) and the induction above. If $\lambda \in L$, its start-copy witness puts $\lambda \in K$ and Rule (S) emits it. □

Theorem 5.3 (exact finite-sample reconstruction)**.** *Let $L \in \mathcal{C}_s^{\mathrm{cf}}$ be nonempty and let G be any CFG for L, with $\mathsf{S}^\sharp$-typed refinement $\widetilde{G}$. If $\mathsf{Wit}(\widetilde{G}) \subseteq K \subseteq L$, then $L(\mathsf{Rec}_s(K)) = L$.*

Proof. Apply Theorem 5.2 to G and Theorem 4.2 to $K \subseteq L$. □

Thus $\mathsf{Wit}(\widetilde{G})$ is characteristic for every nonempty target, including $\{\lambda\}$. For $L = \emptyset$, the empty sample gives $L(\mathsf{Rec}_s(\emptyset)) = \emptyset$. This proves Theorem 3.4(ii).

Corollary 5.4 (Gold identification)**.** *For a fixed surjective semigroup homomorphism $s : \Sigma^+ \to \mathsf{S}$ to a finite semigroup, the learner Learn_s identifies every nonempty language in $\mathcal{C}_s^{\mathrm{cf}}$ from positive data. More precisely, for any CFG G representing the target and its general typed refinement $\widetilde{G}$, once $\mathsf{Wit}(\widetilde{G}) \subseteq K_{n_0}$, among stages $n \geq n_0$ at most one hypothesis change can occur.*

Proof. Let $\mathcal{T}$ be a text for a nonempty L, and choose n_0 with $\mathsf{Wit}(\widetilde{G}) \subseteq K_{n_0}$. Since $w_1 \notin L(H_0) = \emptyset$, the learner rebuilds at stage 1, so for $n \geq 1$ the hypothesis H_n is $\mathsf{Rec}_s(K_m)$ for the most recent rebuild stage $m \leq n$. Sample consistency of that rebuild and the rebuild rule for the later data give $K_n \subseteq L(H_n)$, and soundness gives $L(H_n) \subseteq L$, as $K_m \subseteq L$. If a rebuild occurs at some $n \geq n_0$, Theorem 5.3 gives $L(H_n) = L$, and no later datum triggers a change. Otherwise $H_n = H_{n_0}$ for all $n \geq n_0$, so every word of $\mathcal{T}$ lies in some $K_n \subseteq L(H_n) = L(H_{n_0})$ with $n \geq n_0$; since every word of L occurs in $\mathcal{T}$, $L(H_{n_0}) = L$. In both cases at most one change occurs after stage n_0. □

5.3 Complexity of the reconstruction step

Theorem 5.5 (polynomial-time reconstruction)**.** *For fixed s, $\mathsf{Rec}_s(K)$ is constructible in time polynomial in $\|K\|$.*

Proof. Put $n_K := \|K\|$ and let $\mathcal{O}_K := \{(u, x, v) \in \Sigma^* \times \Sigma^+ \times \Sigma^* : uxv \in K\}$ be the set of observed nonempty factor occurrences. Since $|\mathcal{O}_K| = O(n_K^2)$, there are at most $O(n_K^2)$ factor nonterminals. Extending factors one symbol at a time from each starting position of each sample word computes and caches all their s-values using $O(n_K^2)$ multiplications in the fixed semigroup.

Rule (B) has at most $O(n_K^3)$ candidates, obtained by choosing an observed factor occurrence and a split point. For Rule (U), for each $(u, x, v) \in \mathcal{O}_K$ and each $w \in K$, check whether there is a nonempty word y with $w = uyv$. Such a y, if it exists, is unique; if $s(x) = s(y)$, emit the rule $[\![x]\!] \to [\![y]\!]$. This enumerates all Rule (U) candidates, at most $|\mathcal{O}_K||K| = O(n_K^3)$ of them.

Lexical and start rules are lower-order. Checking, encoding and inserting a candidate take $O(n_K)$ time, and a trie of the explicit rule encodings removes duplicates, so the output is a rule set independent of the enumeration order. Thus $\mathsf{Rec}_s(K)$ is constructible in $O(n_K^4)$ time. □

Each update of Learn_s is also polynomial in the data seen so far, since CFG membership is polynomial and hypotheses and rebuilds are polynomial by Theorem 5.5. Together with sample consistency (Section 4), Theorem 5.3 and Corollary 5.4, this proves Theorem 3.4(i)–(iii); items (iv)–(v) are proved in Sections 7–8. Grammar size alone cannot bound witness lengths: $S_0 \to D_n$, $D_0 \to a$, $D_i \to D_{i-1}D_{i-1}$ has size $O(n)$ but generates $\{a^{2^n}\}$, substitutable for every s [2, Example 2], [14, Example 2.3].

6 Typed Thickness for General Fixed Typings

Definition 6.1 (binary–unit–nullable normal form)**.** *A CFG is in* binary–unit–nullable normal form *(BUNF) if every production is $A \to BC$, $A \to B$, $A \to a$, or $A \to \lambda$.*

Terminal isolation and binarization give a polynomial-size equivalent BUNF grammar; Proposition 7.3 also controls ordinary thickness. Here BUNF bounds typed-rule counts and derivation-tree branching, without restricting Section 5's learning results. Let $G = (V, \Sigma, P, S_0)$

be a reduced BUNF grammar, $\widetilde{G}$ its refinement, and put $N := |V|$ and $N_t := |\widetilde{V} \setminus \{\widetilde{S}\}|$. Define its *typed thickness* by

$$\tau_s^{\mathrm{typ}}(G) := \max\Big(\{\, |\omega(X)| : X \in \widetilde{V} \setminus \{\widetilde{S}\}\,\} \cup \{1\}\Big).$$

It is a presentation-dependent parameter, not an additional assumption on the learner. The typed thickness includes copies of the original start symbol, even if it occurs on a right-hand side or derives λ. In particular, every nonempty $L = L(G)$ satisfies

$$\tau_s^{\mathrm{typ}}(G) \geq \max_{\mu \in \widehat{s}(L)} \min\{\, |w| : w \in L,\ \widehat{s}(w) = \mu\,\},$$

since each nonempty fiber has a retained typed start copy.

Lemma 6.2 (typed-thickness witness bound)**.** *Let G and $\widetilde{G}$ be as above, and put $\tau := \tau_s^{\mathrm{typ}}(G)$. Every non-start typed symbol X satisfies $|u_X| + |v_X| \leq (N_t - 1)\tau$, and each $w \in \mathsf{Wit}(\widetilde{G})$ satisfies $|w| \leq (N_t + 1)\tau$. Let P_{ter}, P_{unit} and P_{bin} be the terminal, unit and binary productions of G. Then*

$$\|\mathsf{Wit}(\widetilde{G})\| \leq \big(N|\mathsf{S}^\sharp| + |P_{\mathrm{ter}}| + |P_{\mathrm{unit}}||\mathsf{S}^\sharp| + |P_{\mathrm{bin}}||\mathsf{S}^\sharp|^2\big)\big((N|\mathsf{S}^\sharp| + 1)\tau + 1\big).$$

For fixed s, this is polynomial in $|G|$ and $\tau_s^{\mathrm{typ}}(G)$.

Proof. In the dependency graph of retained typed symbols include unit and binary dependencies. Each symbol is reachable from a typed copy of S_0; a shortest path has at most $N_t - 1$ edges. Expand off-path siblings of binary rules by canonical yields, each of length at most τ. Unit steps contribute no sibling. Hence the canonical reaching context has length at most $(N_t - 1)\tau$. Each BUNF rule adds at most two canonical yields or one terminal. Empty-type rule witnesses duplicate canonical-yield witnesses and need no extra count. There are at most $N|\mathsf{S}^\sharp|$ typed symbols, $|P_{\mathrm{ter}}|$ terminal typed rules, $|P_{\mathrm{unit}}||\mathsf{S}^\sharp|$ unit typed rules, and $|P_{\mathrm{bin}}||\mathsf{S}^\sharp|^2$ binary typed rules. Summing $|w| + 1$ over the witness set proves the claim. □

Corollary 6.3 (characteristic data via BUNF normalization)**.** *Fix s. Let G_* be any reduced CFG for a nonempty $L(G_*) \in \mathcal{C}_s^{\mathrm{cf}}$, and let G be an equivalent reduced BUNF normalization of size polynomial in $|G_*|$. (If G_* is already BUNF, take $G = G_*$.) Then $\mathsf{Wit}(\widetilde{G})$ is a characteristic sample for Rec_s whose encoded size is polynomial in $|G_*|$ and $\tau_s^{\mathrm{typ}}(G)$.*

Proof. Theorem 5.3 supplies the characteristic-sample property for the original CFG G_*. Apply Lemma 6.2 to its chosen BUNF normalization G, using that $|G|$ is polynomial in $|G_*|$. □

The typed-thickness parameter in Corollary 6.3 belongs to the chosen BUNF normalization, not directly to G_*. For arbitrary fixed s, this does not imply a polynomial bound in the original grammar size and ordinary thickness; indeed, ordinary thickness cannot in general replace typed thickness even for the reconstruction operator.

A bracketed doubling family. Let $\Gamma_{\mathrm{br}} = \{\mathsf{a}, \lfloor, \rfloor, \lceil, \rceil\}$. Let $U_1 = \{1, \zeta\}$ be the two-element semilattice with identity 1 and absorbing element ζ: $1\zeta = \zeta 1 = \zeta^2 = \zeta$. Define the surjective semigroup morphism $s_{\mathrm{U}} : \Gamma_{\mathrm{br}}^+ \to U_1$ by sending $\mathsf{a}, \lfloor, \rfloor$ to 1 and $\lceil, \rceil$ to ζ. Thus $s_{\mathrm{U}}(w) = 1$ exactly for positive words without ceiling brackets, whereas $\widehat{s}_{\mathrm{U}}(\lambda) = \mathbf{1}_\sharp \neq 1$.

For $n \geq 1$, define the reduced CFG G_n by

$$S_0 \to E_n, \qquad E_0 \to \mathsf{a}, \qquad E_i \to \lfloor E_{i-1}E_{i-1}\rfloor \mid \lceil E_{i-1}\rceil \quad (1 \leq i \leq n),$$

and put $L_n = L(G_n)$. Then $|G_n| = O(n)$ and $\tau_{G_n} = 2n+1$. Write $w_0^* = \mathsf{a}$ and $w_i^* = \lfloor w_{i-1}^* w_{i-1}^* \rfloor$. The only word of L_n without ceiling brackets is w_n^*, of length $|w_n^*| = 3 \cdot 2^n - 2$: without unary ceiling-bracket nodes the derivation is the unique complete binary tree of height n.

Proposition 6.4 (refinement can force exponential characteristic data)**.** *Every $n \geq 1$ the following hold.*

(i) *L_n is Clark–Eyraud substitutable, and hence $L_n \in \mathcal{C}^{\mathrm{cf}}_{s_0} \cap \mathcal{C}^{\mathrm{cf}}_{s_{\mathrm{U}}}$.*

(ii) *L_n has a characteristic sample for Rec_{s_0} of encoded size polynomial in n.*

(iii) *Every finite $K \subseteq L_n$ such that $w_n^* \in L(\mathsf{Rec}_{s_{\mathrm{U}}}(K))$ contains w_n^*. Consequently, every characteristic sample $\mathcal{F}$ of L_n for $\mathsf{Rec}_{s_{\mathrm{U}}}$ satisfies $\|\mathcal{F}\| \geq 3 \cdot 2^n - 1$.*

(iv) *Put $L_n^- := L_n \setminus \{w_n^*\}$. Then $L_n^- \in \mathcal{C}^{\mathrm{cf}}_{s_{\mathrm{U}}}$. For any set-driven reconstruction operator Op with characteristic samples for both L_n and L_n^-, every characteristic sample $\mathcal{F}$ of L_n for Op contains w_n^* and hence satisfies $\|\mathcal{F}\| \geq 3 \cdot 2^n - 1$. In particular, this holds for every set-driven operator having characteristic samples for the entire fixed-s_{U} class.*

In particular, for the fixed typing s_{U} no bound on characteristic-data size for $\mathsf{Rec}_{s_{\mathrm{U}}}$ polynomial in source grammar size and ordinary thickness holds over all reduced CFG presentations of targets in $\mathcal{C}^{\mathrm{cf}}_{s_{\mathrm{U}}}$.

Proof. The substitutability assertion (i) is proved in Appendix C. For (ii), an explicit sample suffices. Set $u_0 = \mathsf{a}$, $u_i = \lceil u_{i-1} \rceil$, and

$$\mathcal{F}_n = \{u_n\} \cup \{\lceil^{n-i} \lfloor u_{i-1} u_{i-1} \rfloor \rceil^{n-i} : 1 \leq i \leq n\}.$$

For each i, the two sample words provide a common context for u_i and $\lfloor u_{i-1} u_{i-1} \rfloor$; Rules (U) and (B) therefore recover both E_i alternatives, so $L_n \subseteq L(\mathsf{Rec}_{s_0}(\mathcal{F}_n))$. For any $\mathcal{F}_n \subseteq K \subseteq L_n$, monotonicity applied to $\mathcal{F}_n \subseteq K$ gives $L_n \subseteq L(\mathsf{Rec}_{s_0}(K))$, while soundness (Theorem 4.2) applied to $K \subseteq L_n$ gives $L(\mathsf{Rec}_{s_0}(K)) \subseteq L_n$. Thus $\mathcal{F}_n$ is characteristic, with $|\mathcal{F}_n| = n + 1$ and $\|\mathcal{F}_n\| = 3n^2 + 4n + 2$.

For (iii)–(iv), the deleted-word target is the type-fiber filter $L_n^- = L_n \cap \widehat{s}_{\mathrm{U}}^{-1}(\{\zeta\})$. It belongs to $\mathcal{C}^{\mathrm{cf}}_{s_{\mathrm{U}}}$: equal s_{U}-types of nonempty factors remain equal $\widehat{s}_{\mathrm{U}}$-types inside any common outer context. Thus intersection with a union of $\widehat{s}_{\mathrm{U}}$-fibers preserves relative substitutability, and the intersection is context-free by regular filtering.

For (iii), if $K \subseteq L_n$ omits w_n^*, then $K \subseteq L_n^-$. Soundness (Theorem 4.2) gives $L(\mathsf{Rec}_{s_{\mathrm{U}}}(K)) \subseteq L_n^-$, so w_n^* cannot be generated. Hence every characteristic sample $\mathcal{F}$ contains w_n^* and satisfies $\|\mathcal{F}\| \geq |w_n^*| + 1 = 3 \cdot 2^n - 1$. Together with $|G_n|, \tau_{G_n} = O(n)$, this proves the final assertion.

For (iv), let $\mathcal{F}$ be characteristic for L_n under Op and suppose $w_n^* \notin \mathcal{F}$. Take a characteristic sample $\mathcal{F}'$ for L_n^-. Then $\mathcal{F} \cup \mathcal{F}' \subseteq L_n^- \subsetneq L_n$, so the same output $\mathsf{Op}(\mathcal{F} \cup \mathcal{F}')$ would have to generate both languages, a contradiction.

For $n \geq 2$, $L_n^- \notin \mathcal{C}^{\mathrm{cf}}_{s_0}$: u_{n-1} and w_{n-1}^* share the context $(\lfloor, u_{n-1} \rfloor)$ in L_n^-, but only the former admits $(\lfloor w_{n-1}^*, \rfloor)$. Thus (iv) does not conflict with the s_0 upper bound in (ii). □

The contradiction in part (iv) follows the classical distinguishing-pair principle using a word in the symmetric difference of two targets; compare [4, Theorem 15.1.4]. The fixed-s distinguishing pair L_n, L_n^- and its exponential witness length are specific to the present construction.

Corollary 6.5 (presentation-independent typed-thickness gap)**.** *Every reduced BUNF grammar $G = (V, \Gamma_{\mathrm{br}}, P, S_0)$ for L_n satisfies*

$$\boxed{\tau^{\mathrm{typ}}_{s_{\mathrm{U}}}(G) \geq 3 \cdot 2^n - 2.}$$

This holds regardless of grammar size. Conversely, L_n has a reduced BUNF presentation with both size and ordinary thickness polynomial in n.

Proof. The positive fiber of type 1 is $L_n \cap \widehat{s}_{\mathrm{U}}^{-1}(\{1\}) = \{w_n^*\}$, with $1 \neq \mathbf{1}_\sharp$. The fiber-minimum bound above therefore gives $\tau^{\mathrm{typ}}_{s_{\mathrm{U}}}(G) \geq |w_n^*| = 3 \cdot 2^n - 2$ for every such G. For the ordinary-thickness upper construction, apply Proposition 7.3 to G_n. □

7 Polynomial Thick Data Beyond Fixed Windows

Yoshinaka [3, Theorem 1] already established polynomial thick-data learnability for fixed (k,ℓ)-substitutable context-free languages using (k,ℓ)-Greibach normal form and characteristic-set bounds [3, Lemmas 2 and 7]. Here we instead bound the shortest yields of the typed refinement used by Rec_s. One argument covers both fixed windows and typings that additionally track finite-group values. The corresponding batch bounds establish identification in polynomial time and thick data (IPTtD), in the set-driven sense of [3] and [14, Definition 2.12]. They do not assert the stronger incremental stabilization requirement of [14, Definition 2.5]. We claim no improvement in Yoshinaka's polynomial degree. For general fixed s, ordinary thickness need not give a polynomial characteristic-data bound for Rec_s (Proposition 6.4).

Fix $k,\ell \geq 0$ and a morphism $g : \Sigma^* \to \mathsf{H}$ into a finite group. In products with $s_{k,\ell}$, the group component is restricted to Σ^+. Put $m := k + \ell$.

Lemma 7.1 (window–group typed-yield bound)**.** *Let $G = (V, \Sigma, P, S_0)$ be a reduced BUNF grammar, $N := |V| \geq 1$, $m := k + \ell$ and $q := |\mathsf{H}|$. If $\ker\big(s_{k,\ell} \times (g|_{\Sigma^+})\big) \subseteq \ker s$, then*

$$\tau_s^{\mathrm{typ}}(G) \leq m + 1 + [1 + 2(m+q)(m+2)Nq](\tau_G + 1).$$

This includes nullable and unit productions, including productions whose right-hand sides contain S_0.

Proof. See Appendix A, where a nonemptiness anchor and derivation-size tie break handle the additional BUNF cases. □

Theorem 7.2 (polynomial thick data under window–group refinement)**.** *Fix s, Σ and suppose $\ker\big(s_{k,\ell} \times (g|_{\Sigma^+})\big) \subseteq \ker s$ for some fixed k,ℓ and finite-group morphism g. Every nonempty $L \in \mathcal{C}_s^{\mathrm{cf}}$ represented by a reduced BUNF grammar G has a characteristic sample $\mathsf{Wit}(\widetilde{G})$ for Rec_s of encoded size polynomial in $|G|$ and $\tau_G + 1$.*

Proof. Combine Lemma 7.1 and Corollary 6.3. □

Theorem 7.2 includes fixed windows when H is trivial. For arbitrary CFG presentations we need only terminal isolation and binarization.

Proposition 7.3 (polynomial thickness-preserving BUNF normalization)**.** *There are fixed polynomials q_{size} and q_{thick} such that every reduced CFG G_* generating a nonempty language can be converted in polynomial time into an equivalent reduced BUNF grammar G with $|G| \leq q_{\mathrm{size}}(|G_*|)$ and $\tau_G \leq q_{\mathrm{thick}}(|G_*|, \tau_{G_*} + 1)$. The construction retains nullable and unit productions.*

See Appendix A.2.

Lemma 7.4 (locally trivial–group divisors and window refinements)**.** *Let $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ be a finite-semigroup typing on a nonempty alphabet. The following are equivalent:*

(i) *S divides a product $\mathsf{T} \times \mathsf{H}$, where T is a finite locally trivial semigroup and H is a finite group;*

(ii) *for some $k,\ell \geq 0$ and a monoid morphism $g : \Sigma^* \to \mathsf{H}$ into a finite group H,*

$$\ker\big(s_{k,\ell} \times (g|_{\Sigma^+})\big) \subseteq \ker s.$$

Proof. For (i), choose a subsemigroup $V \subseteq \mathsf{T} \times \mathsf{H}$ and a surjective morphism $\pi : V \to \mathsf{S}$. Lift the letter images $s(a)$ to V, obtaining $\varphi : \Sigma^+ \to V$ with $\pi\varphi = s$. The factor projections are $\sigma : \Sigma^+ \to \mathsf{T}$ and $g_+ : \Sigma^+ \to \mathsf{H}$. The latter extends to $g : \Sigma^* \to \mathsf{H}$ with $g(\lambda) = 1_{\mathsf{H}}$. Since $\sigma(\Sigma^+)$ is locally trivial, Proposition 3.5 supplies k,ℓ with $\ker s_{k,\ell} \subseteq \ker \sigma$. Together with g_+ this implies (ii).

Conversely (ii) induces a surjective semigroup morphism

$$(s_{k,\ell} \times g)(\Sigma^+) \longrightarrow \mathsf{S}, \qquad (s_{k,\ell}(w), g(w)) \longmapsto s(w).$$

The domain is a subsemigroup of $W_{k,\ell} \times \mathsf{H}$, with $W_{k,\ell}$ locally trivial by Proposition 3.5; hence (i). □

Corollary 7.5 (ordinary-thickness bounds for locally trivial–group typings)**.** *Fix $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$. Suppose S is a homomorphic image of a subsemigroup of the direct product of a finite locally trivial semigroup and a finite group. Then every nonempty $L \in \mathcal{C}_s^{\mathrm{cf}}$ represented by a reduced CFG G_* has characteristic data for Rec_s of encoded size polynomial in $|G_*|$ and $\tau_{G_*} + 1$. This includes all locally trivial positive-image typings and all finite-group typings.*

Proof. By Lemma 7.4, Theorem 7.2 applies to the BUNF grammar given by Proposition 7.3; compose the polynomials. □

Let **LI** and **Grp** be the pseudovarieties of finite locally trivial semigroups and of finite groups. As both are closed under finite direct products, their join $\mathbf{LI} \vee \mathbf{Grp}$ consists exactly of the divisors (homomorphic images of subsemigroups) of products $\mathsf{S} \times \mathsf{G}$ with $\mathsf{S} \in \mathbf{LI}$ and $\mathsf{G} \in \mathbf{Grp}$; such joins were studied by Costa [19, 20]. The hypothesis of Corollary 7.5 thus reads $s(\Sigma^+) \in \mathbf{LI} \vee \mathbf{Grp}$.

Lemma 7.6 (the two-element semilattice lies outside the window–group join)**.** $U_1 \notin \mathbf{LI} \vee \mathbf{Grp}$.

Proof. Suppose there were a subsemigroup $\mathsf{T} \subseteq \mathsf{S} \times \mathsf{G}$ and a surjective morphism $\pi : \mathsf{T} \to U_1$, with $\mathsf{S} \in \mathbf{LI}$ and G a finite group. Choose idempotents $e \in \pi^{-1}(1)$ and $f \in \pi^{-1}(\zeta)$ (Section 2.2). Every idempotent of $\mathsf{S} \times \mathsf{G}$ has group component 1_{G}, so the S-components of e, f are idempotents. Local triviality of S gives $efe = e$. Applying π would yield $1 \cdot \zeta \cdot 1 = 1$, contradicting $\zeta \neq 1$. □

Remark 7.7. The positive image of s_{U} is U_1, which lies outside $\mathbf{LI} \vee \mathbf{Grp}$ by Lemma 7.6. Proposition 6.4 shows that the ordinary-thickness characteristic-data bound in Corollary 7.5 can fail for Rec_s when its algebraic hypothesis is removed. This does not establish that the hypothesis is necessary for every fixed typing. Proposition 6.4(iv) does give a reconstruction-operator-independent lower bound on characteristic-sample size for the distinguishing target pair, but not a lower bound on the running time or convergence stage of arbitrary Gold learners.

These polynomial bounds may depend on the fixed typing and alphabet; no uniform bound for variable windows or groups is claimed.

8 The Linear Subclass

Linear languages are exactly the languages accepted by one-turn pushdown automata [21, 22]. The normalization below gives the grammatical counterpart: every successful derivation has one continuing nonterminal path, which we call its *spine*.

Proposition 8.1 (linear-spine BUNF normalization)**.** *There is a fixed polynomial q_{lin} such that every linear CFG G generating a nonempty language can be converted in polynomial time into an equivalent reduced BUNF grammar $G_0 = (V_0, \Sigma, P_0, S_0)$ with $|G_0| \le q_{\mathrm{lin}}(|G|)$ in which every binary production has exactly one fresh terminal-wrapper child W_a with $W_a \to a$ and exactly one other nonterminal child.*

The construction is given in Appendix B. Let $\widetilde{G}_0$ be its typed refinement and $N_t := |\widetilde{V}_0 \setminus \{\widetilde{S}\}| \le |V_0|\,|\mathsf{S}^\sharp|$.

Lemma 8.2 (short canonical yields)**.** *Every typed non-start symbol X of $\widetilde{G}_0$ satisfies $|\omega(X)| \le N_t$. Hence $\tau_s^{\mathrm{typ}}(G_0) \le N_t$.*

Proof. The *continuing spine* follows the unique non-wrapper child of each binary rule. The wrapper child generates exactly one terminal and never continues the spine. Unit rules add no terminals; lexical and nullable rules end the spine. Choose a derivation of $\omega(X)$ with the fewest nonterminal occurrences. A repeated *typed* spine symbol permits grafting its lower subderivation at the upper occurrence, preserving a valid typed derivation. A segment containing a binary rule would strictly shorten $\omega(X)$; a unit-only segment would decrease the number of nodes. The former contradicts the minimal length of $\omega(X)$; the latter contradicts the choice of a derivation with the fewest nonterminal occurrences. At most N_t spine symbols occur, and each contributes at most one terminal, including the final lexical rule. Therefore $|\omega(X)| \le N_t$. Wrapper symbols also satisfy this estimate. □

Theorem 8.3 (polynomial characteristic data for linear targets)**.** *For fixed s, every $L \in \mathcal{C}_s^{\mathrm{lin}}$ represented by a linear CFG G has a characteristic sample for Rec_s whose $\|\cdot\|$-size is polynomial in $|G|$.*

Proof. For $L = \emptyset$ and $L = \{\lambda\}$ use the samples from Section 5. Otherwise take G_0 from Proposition 8.1. Lemma 8.2 gives $\tau_s^{\mathrm{typ}}(G_0) \le N_t \le |V_0|\,|\mathsf{S}^\sharp|$, and Corollary 6.3 yields a characteristic sample polynomial in $|G_0| \le q_{\mathrm{lin}}(|G|)$. More explicitly, the bound in Lemma 6.2 gives $\|\mathsf{Wit}(\widetilde{G}_0)\| = O(|G_0|^3|\mathsf{S}^\sharp|^4)$; the polynomial dependence on the original $|G|$ also uses the normalization bound q_{lin}. □

Together with Theorem 5.5, this gives polynomial-time batch reconstruction and polynomial characteristic data relative to a chosen linear CFG for targets in $\mathcal{C}_s^{\mathrm{lin}}$. This does not contradict de la Higuera's negative result for the unrestricted class of linear languages [15, Theorem 2]: our theorem concerns only the fixed-s substitutable subclass. Moreover, no incremental polynomial-data guarantee for Learn_s is asserted.

A nonregular linear fixed-s example. The preceding theorem is not confined to regular targets, nor to the Clark–Eyraud or fixed-window classes. Define

$$L_{\pm,e} := \{a^{2m}cb^{2m} : m \ge 0\} \cup \{a^{2m+1}db^{2m+1} : m \ge 0\} \cup \{a^n eb^n : n \ge 0\}.$$

It is generated by the linear grammar

$$S \to U_{\mathrm{ev}} \mid U_e, \quad U_{\mathrm{ev}} \to c \mid aU_{\mathrm{odd}}b, \quad U_{\mathrm{odd}} \to d \mid aU_{\mathrm{ev}}b, \quad U_e \to e \mid aU_eb.$$

The subscripts ev and odd track the parity of enclosing a, b-pairs from the start, not the length parity of the corresponding nonterminal's own yield.

Call c, d, e the *center symbols*.

Proposition 8.4. *The language $L_{\pm,e}$ is linear, nonregular, and belongs to $\mathcal{C}_s^{\mathrm{lin}}$ for some homomorphism s into a finite semigroup. It is neither Clark–Eyraud substitutable nor (k, ℓ)-substitutable for any fixed $k, \ell \ge 0$.*

Proof. Linearity follows from the displayed grammar. To establish finite-typed substitutability, put

$$L_{\mathrm{all}} := \{a^n \xi b^n : n \ge 0,\ \xi \in \{c, d, e\}\}, \qquad R_{\mathrm{reg}} := (aa)^*c(bb)^* \cup a(aa)^*d\,b(bb)^* \cup a^*eb^*.$$

The linear language L_{all} is Clark–Eyraud substitutable under the nonempty-factor convention: a shared context uniquely determines a center-free factor, while the distribution of a factor $a^i\xi b^j$

containing one center depends only on $i-j$. Thus $L_{\mathrm{all}} \in \mathcal{C}^{\mathrm{lin}}_{s_0}$ for the trivial typing s_0. Since R_{reg} is regular and $L_{\pm,e} = L_{\mathrm{all}} \cap R_{\mathrm{reg}}$, Proposition 3.2(ii), together with closure of linear languages under regular intersection, gives $L_{\pm,e} \in \mathcal{C}^{\mathrm{lin}}_{s_0\times(g|_{\Sigma^+})}$ for a finite-monoid morphism $g : \Sigma^* \to M$ recognizing R_{reg}.

Moreover, $L_{\pm,e} \cap a^*eb^* = \{a^neb^n : n \geq 0\}$ proves nonregularity. The factors c, e share the empty context, but $(a,b) \in \mathcal{D}_{L_{\pm,e}}(e) \setminus \mathcal{D}_{L_{\pm,e}}(c)$; hence $L_{\pm,e}$ is not Clark–Eyraud substitutable. Finally, fix $k, \ell \geq 0$ and put $m_0 := \max\{1, k+\ell\}$, $w_e := a^keb^\ell$, and $w_c := a^kcb^\ell$. The factors w_e, w_c have the same length-k prefix and length-ℓ suffix and share the context $(a^{2m_0-k}, b^{2m_0-\ell})$, whereas $(a^{2m_0+1-k}, b^{2m_0+1-\ell}) \in \mathcal{D}_{L_{\pm,e}}(w_e) \setminus \mathcal{D}_{L_{\pm,e}}(w_c)$. The fixed-window counterexample criterion gives $L_{\pm,e} \notin \mathsf{KL}$. □

A concrete four-element typing. The regular-filter argument establishes the required existence of a finite typing. In fact, a four-element typing already suffices. Let $M_{\pm,e} := \{1, \kappa_{cd}, \kappa_e, 0\}$ be the monoid with identity 1 and $\mu\nu = 0$ for all $\mu, \nu \in \{\kappa_{cd}, \kappa_e, 0\}$, and define $s_{\pm,e} : \{a,b,c,d,e\}^+ \to M_{\pm,e}$ by

$$s_{\pm,e}(a) = s_{\pm,e}(b) = 1, \qquad s_{\pm,e}(c) = s_{\pm,e}(d) = \kappa_{cd}, \qquad s_{\pm,e}(e) = \kappa_e. \tag{3}$$

Then $L_{\pm,e}$ is $\sim_{s_{\pm,e}}$-substitutable. Indeed, two center-free nonempty factors sharing a context must coincide. The distribution of a^ieb^j depends only on $i-j$, while that of $a^i\xi b^j$ for $\xi \in \{c,d\}$ depends only on $(i-j, \mathrm{par}(\xi) - i \bmod 2)$, where $\mathrm{par}(c) = 0$ and $\mathrm{par}(d) = 1$. A shared context forces these invariants to agree, and equal $s_{\pm,e}$-types distinguish e-factors from c/d-factors; thus equal-type factors with a common context have the same distribution.

Thus Theorem 8.3 applies to a nonregular linear language outside KL. The displayed grammar is even-linear, so no separation from Takada's control-set framework is claimed.

Finite-state control induced by yield typing. A single finite-state construction covers both the original typing s and its completion $\widehat{s}$. Let $t : \Sigma^* \to T$ be any finite-monoid morphism, and label each production of a linear grammar $G = (V, \Sigma, P, S)$ distinctly over $\mathsf{ProdLab}$. Write $A \Rightarrow_G^{\alpha} w$ for a derivation with production-label word α. For $\mathsf{Acc}_{\mathrm{ctrl}} \subseteq T$, put

$$\mathsf{Ctrl}_{t,\mathsf{Acc}_{\mathrm{ctrl}}}(G) := \{\alpha \in \mathsf{ProdLab}^* : S \Rightarrow_G^{\alpha} w \text{ for some } w \in \Sigma^*,\ t(w) \in \mathsf{Acc}_{\mathrm{ctrl}}\}.$$

This is regular: an automaton with states $V \times T \times T$ and an accepting state starts at $(S, 1_T, 1_T)$, updates $(A, \mu_L, \mu_R) \mapsto (B, \mu_L t(u), t(v)\mu_R)$ for each $A \to uBv$, and accepts a terminal rule $A \to w$ iff $\mu_L t(w) \mu_R \in \mathsf{Acc}_{\mathrm{ctrl}}$. The invariant is $(A, t(u), t(v))$ for a sentential form uAv; thus the controlled grammar generates $L(G) \cap t^{-1}(\mathsf{Acc}_{\mathrm{ctrl}})$.

The semigroup typing s is used through its monoid extension: taking $t = \widehat{s}$, start symbol A, and $\mathsf{Acc}_{\mathrm{ctrl}} = \{\mu\}$ realizes exactly $L_G(A) \cap \widehat{s}^{-1}(\{\mu\})$, the fiber in Proposition 5.1, including the nullable type $\mathbf{1}_\sharp$. This is a finite-state bridge to Takada's regular control languages [11]; we do not claim arbitrary Takada control sets arise from yield typings.

9 Expressiveness and Boundaries

We now locate the framework between the fixed-window hierarchy and broader context-free families.

9.1 A nonlinear target inside RS

Let $\Delta := \{a^nb^n : n \geq 0\}$ and consider Δ^*. For $z \in \{a,b\}^*$ put $\beta(z) := |z|_a - |z|_b$; the *height* after reading a prefix p is $\beta(p)$. For any word x, define its *admissible entry-height set* by

$$\mathcal{Q}(x) := \{q \in \mathbb{Z}_{\geq 0} : q + \beta(x) \geq 0, \quad a^qxb^{q+\beta(x)} \in \Delta^*\}.$$

A useful local characterization is that $w \in \Delta^*$ if and only if all prefix heights of w are nonnegative, $\beta(w) = 0$, and immediately after the b in every occurrence of ba the height is zero. Indeed, the ba boundaries split w into blocks of the form $a^i b^j$; the stated conditions force each block to start and end at height zero and hence to have $i = j$. Conversely, every concatenation of such blocks has the stated properties. Applying this characterization to the canonical completion $a^q x b^{q+\beta(x)}$ (whose boundaries with x introduce no new ba factor) shows that $\mathcal{Q}(x)$ is exactly the set of heights $\beta(u)$ at which x occurs in a word $uxv \in \Delta^*$. For the converse direction, reading x from height $\beta(u)$ in such a word preserves all internal nonnegativity and ba-boundary conditions; its canonical completion therefore belongs to Δ^*.

For example, $\mathcal{Q}(ab) = \mathbb{Z}_{\geq 0}$, $\mathcal{Q}(aa) = \mathbb{Z}_{\geq 0}$, $\mathcal{Q}(\lambda) = \mathbb{Z}_{\geq 0}$, $\mathcal{Q}(bb) = \mathbb{Z}_{\geq 2}$, $\mathcal{Q}(ba) = \{1\}$, $\mathcal{Q}(bbaa) = \{2\}$, and $\mathcal{Q}(baaba) = \varnothing$.

The canonical completion gives the membership criterion

$$w \in \Delta^* \iff \beta(w) = 0 \text{ and } 0 \in \mathcal{Q}(w). \tag{$\star$}$$

Let $\mathrm{fst}(w), \mathrm{lst}(w) \in \{\bot, a, b\}$ be the first and last symbols, with $\bot$ on λ, and let $\mathrm{ba}(w)$ record whether w contains ba. Put $h_\star(w) := (\mathrm{fst}(w), \mathrm{lst}(w), \mathrm{ba}(w))$. The summary of a concatenation is determined by the summaries of its factors: the endpoints concatenate in the evident way and a new ba can arise only across a boundary $b|a$. Hence the finite image $M_\star := h_\star(\{a, b\}^*)$ carries the product $h_\star(x)h_\star(y) := h_\star(xy)$, making $h_\star$ a finite-monoid homomorphism. Write $s_\star := h_\star|_{\{a,b\}^+}$ for the positive-word learning typing.

Appending a word on the right cannot enlarge the admissible entry-height set; more precisely, the following concatenation identity holds:[2]

$$\begin{aligned} \mathcal{Q}(st) = \mathcal{Q}(s) \cap \{q \in \mathbb{Z} : q + \beta(s) \in \mathcal{Q}(t)\} \\ \cap \begin{cases} \{-\beta(s)\}, & (\mathrm{lst}(s), \mathrm{fst}(t)) = (b, a), \\ \mathbb{Z}, & \text{otherwise.} \end{cases} \end{aligned} \tag{$\star\star$}$$

Proposition 9.1 (a nonlinear fixed-$s_\star$ target)**.** *The language Δ^* is deterministic context-free, nonregular, and nonlinear, and it is $\sim_{s_\star}$-substitutable. Moreover it is not (k, ℓ)-substitutable for any fixed $k, \ell \geq 0$. Hence $\Delta^* \in \mathsf{RS}_{s_\star}(\{a, b\}) \cap \mathrm{CFL}$ and $\Delta^* \in \mathsf{RS}(\{a, b\}) \setminus \mathsf{KL}(\{a, b\})$.*

Proof. The grammar $S \to TS \mid \lambda$, $T \to aTb \mid \lambda$ generates Δ^*. It is deterministic context-free: a DPDA pushes while reading the a-part of a factor $a^n b^n$, pops during its b-part, and permits a new such factor only after the stack returns to its bottom marker. Nonregularity follows from $\Delta^* \cap a^*b^* = \Delta$, and nonlinearity from closure of linear languages under regular intersection together with $\Delta^* \cap a^*b^*a^*b^* = \Delta\Delta$. The nonlinearity of $\Delta\Delta$ is explicitly recorded by Autebert–Berstel–Boasson [22, Section 6.1, before Theorem 6.1].

For substitutability, let nonempty x, y have equal $s_\star$-type and a common context (u, v). By the preceding characterization of $\mathcal{Q}$ and additivity of β, the memberships $uxv, uyv \in \Delta^*$ imply $\beta(x) = \beta(y)$ and $\beta(u) \in \mathcal{Q}(x) \cap \mathcal{Q}(y)$. Since $s_\star(x) = s_\star(y)$, either both contain ba or neither does.

In the former case, write $x = rbat$ with $r, t \in \{a, b\}^*$. In any word of Δ^*, the height immediately after the b in each occurrence of ba is 0. Hence every $q \in \mathcal{Q}(x)$ satisfies $q + \beta(rb) = 0$, so $\mathcal{Q}(x)$ has at most one element; the same holds for $\mathcal{Q}(y)$. Since $\beta(u) \in \mathcal{Q}(x) \cap \mathcal{Q}(y)$, we obtain $\mathcal{Q}(x) = \{\beta(u)\} = \mathcal{Q}(y)$.

In the latter case, $x, y \in a^*b^*$, and for each $z \in \{x, y\}$ we have $\mathcal{Q}(z) = \{q \in \mathbb{Z} : q \geq -\min\{0, \beta(z)\}\}$. Since we have already shown $\beta(x) = \beta(y)$, it follows immediately that $\mathcal{Q}(x) = \mathcal{Q}(y)$.

[2] Indeed, the conditions for reading st from height q split into those within s, namely $q \in \mathcal{Q}(s)$; those within t, namely $q + \beta(s) \in \mathcal{Q}(t)$; and the condition at the boundary $s|t$. Only when this boundary is $b|a$ must the height immediately after that b be 0, requiring $q + \beta(s) = 0$. These conditions are also sufficient, and the identity follows.

Thus $\beta(x) = \beta(y)$ and $\mathcal{Q}(x) = \mathcal{Q}(y)$. Also, $\mathrm{fst}(x) = \mathrm{fst}(y)$ and $\mathrm{lst}(x) = \mathrm{lst}(y)$ by equality of $s_\star$-types. Applying $(\star\star)$ twice gives, for all u', v', $\beta(u'xv') = \beta(u'yv')$ and $\mathcal{Q}(u'xv') = \mathcal{Q}(u'yv')$. By $(\star)$, $u'xv' \in \Delta^*$ iff $u'yv' \in \Delta^*$. Thus $\mathcal{D}_{\Delta^*}(x) = \mathcal{D}_{\Delta^*}(y)$.

For the fixed-window separation, fix k, ℓ and let $m := \max\{1, k + \ell\}$, $x := a^m b^m$, and $y := a^m b^m a^m b^m$. They have the same length-k prefix and length-ℓ suffix and share the empty context. Moreover $axb \in \Delta^*$, whereas the central ba in ayb has height 1, so $ayb \notin \Delta^*$. The fixed-window counterexample criterion therefore excludes every (k, ℓ). □

9.2 A capped regular counter family

Yoshinaka's regular language $L_{\mathrm{Yreg}} = ae^*ce^*a \cup ae^*de^*a \cup be^*ce^*b$ already gives a fixed-window separation [3, Proposition 1]; capped counters contrast with the uncapped obstruction below.

Let $\Sigma_c := \{\uparrow, \downarrow, \#\}$. For $w = \delta_1 \cdots \delta_n \in \Sigma_c^*$, let $\mathrm{ht}_i(w)$ be the height after i letters, with $\mathrm{ht}_0(w) = 0$, changing by $+1$ on $\uparrow$, -1 on $\downarrow$, and resetting to 0 on $\#$. For $\rho \geq 1$, put

$$\mathrm{CTR}_\rho := \{w \in \Sigma_c^* : 0 \leq \mathrm{ht}_i(w) \leq \rho \text{ for all } 0 \leq i \leq |w|\}.$$

For every $\rho \geq 1$, CTR_ρ is regular: a deterministic finite automaton records heights $0, \ldots, \rho$, uses one sink outside the range, and resets to 0 on $\#$. Hence $\mathrm{CTR}_\rho \in \mathsf{RS}(\Sigma_c)$ by Proposition 3.1.

Theorem 9.2. *For every $\rho \geq 2$, CTR_ρ is not (k, ℓ)-substitutable for any $k, \ell \geq 0$.*

Proof. Fix k, ℓ and put $x := (\uparrow\downarrow)^k \uparrow^{\rho-1} \#^\ell$ and $y := (\uparrow\downarrow)^k \uparrow^\rho \#^\ell$. They are long enough for the fixed-window criterion, have the same length-k prefix and length-ℓ suffix, and both lie in CTR_ρ, so they share the empty context. Yet $\uparrow x \in \mathrm{CTR}_\rho$ while $\uparrow y \notin \mathrm{CTR}_\rho$, the latter reaching height $\rho + 1$. Hence their distributions differ, and the criterion applies. □

The bound $\rho \geq 2$ is sharp[3]. Let $h_\rho : \Sigma_c^* \to M_\rho$ be the DFA transition-monoid morphism recognizing CTR_ρ. Its positive restriction $s_\rho := h_\rho|_{\Sigma_c^+}$ is a finite-semigroup typing witnessing $\mathrm{CTR}_\rho \in \mathcal{C}^{\mathrm{cf}}_{s_\rho}$; only $\rho \geq 2$ lies outside the fixed-window hierarchy.

9.3 Natural deterministic context-free languages outside RS

Lemma 9.3 (finite-semigroup obstruction)**.** *Let $L \subseteq \Sigma^*$. Suppose an infinite $\Xi \subseteq \Sigma^+$ satisfies, for all distinct $x, y \in \Xi$, $\mathcal{D}_L(x) \cap \mathcal{D}_L(y) \neq \emptyset$ and $\mathcal{D}_L(x) \neq \mathcal{D}_L(y)$. Then $L \notin \mathsf{RS}(\Sigma)$.*

Proof. Any finite s identifies two elements of Ξ; their common context and unequal distributions contradict $\sim_s$-substitutability. □

Removing the upper cap gives the deterministic context-free language

$$\mathrm{CTR} := \{w \in \Sigma_c^* : \mathrm{ht}_i(w) \geq 0 \text{ for all } 0 \leq i \leq |w|\},$$

accepted by the evident deterministic pushdown counter with reset.

Corollary 9.4 (uncapped counter obstruction)**.** $\mathrm{CTR} \notin \mathsf{RS}(\Sigma_c)$.

Proof. Use $\Xi = \{\uparrow^i : i \geq 1\}$. For $i < j$, $(\lambda, \downarrow^i)$ is common, while $(\lambda, \downarrow^j)$ is a context only of $\uparrow^j$. □

Define the one-bracket Dyck language by

$$D_1 := \{w \in \{a, b\}^* : |w'|_a \geq |w'|_b \text{ for every prefix } w',\ |w|_a = |w|_b\}.$$

It is a standard deterministic context-free language [24]. Here D_1 denotes the whole Dyck language, written D_1^* in the convention of Autebert–Berstel–Boasson [22].

[3]Indeed, CTR_1 is $(1, 1)$-substitutable. Membership is characterized by the initial symbol not being $\downarrow$ and the absence of the length-2 factors $\uparrow\uparrow$, $\downarrow\downarrow$, and $\# \downarrow$. Thus, for nonempty factors occurring in words of CTR_1 with a common context, equality of their first and last symbols ensures equal membership under every context.

Corollary 9.5. $D_1 \notin \mathsf{RS}(\{a,b\})$.

Proof. Use $\Xi = \{b^i a^i : i \geq 1\}$. For $i < j$, (a^j, b^j) is common, whereas (a^i, b^i) is a context only of $b^i a^i$; for $b^j a^j$ the prefix $a^i b^j$ underflows. □

Thus unbounded stack behavior alone is not the obstruction: the nonregular one-turn language $L_{\pm,e}$ lies in RS, but an infinite family of mutually comparable factors with distinct contextual behavior cannot be separated by any finite s.

Lemma 9.6 (fixed-word quotient)**.** *If L is $\sim_s$-substitutable and $z \in \Sigma^*$, then $L/z := \{w : wz \in L\}$ is $\sim_s$-substitutable.*

Proof. A common context (u, v) in L/z becomes (u, vz) in L; equality of distributions in L then transfers every context (s, t) by appending z. □

The language generated by $S \to aSS \mid b$ is the *Łukasiewicz language*. Yoshinaka [3, Section 3] already observed that this language is not (k,ℓ)-substitutable for any fixed k, ℓ. In the notation used here, $L_{\mathrm{Luk}} = D_1 b$ [22], and it is deterministic context-free [24]. The quotient argument gives the stronger conclusion that it lies outside the present finite-typing union: if $L_{\mathrm{Luk}} \in \mathsf{RS}(\{a,b\})$, then Lemma 9.6 would put $D_1 = L_{\mathrm{Luk}}/b$ in RS, contradicting Corollary 9.5.

9.4 Comparison with Clark's congruential family

Clark [25] allows a finite initial set $I \subseteq V$; write $L_G[I] := \bigcup_{A \in I} L_G(A)$. Such a grammar is *congruential* if each $L_G(A)$ lies in one syntactic-congruence class of $L_G[I]$. Let $\mathsf{CONG}(\Gamma)$ be the languages over Γ generated by such grammars. Clark already places substitutable and (k,ℓ)-substitutable languages in this family; the following extends the containment to every fixed-s slice.

Proposition 9.7 (comparison with Clark's congruential family)**.** *For every finite alphabet Γ and finite-semigroup typing $s : \Gamma^+ \to \mathsf{S}$, $\mathcal{C}_s^{\mathrm{cf}} \subseteq \mathsf{CONG}(\Gamma)$. Consequently $\mathsf{RS}(\Gamma) \cap \mathrm{CFL} \subseteq \mathsf{CONG}(\Gamma)$, and for $\Gamma = \{a,b\}$ the inclusion is proper.*

Proof. For nonempty $L \in \mathcal{C}_s^{\mathrm{cf}}$, take any CFG for L and its trimmed $\mathsf{S}^\sharp$-typed refinement $\widetilde{G}$. The yields of each retained typed symbol A_μ have equal $\widehat{s}$-type and share a terminal reaching context. Lemma 2.2 implies that they lie in one syntactic-congruence class. This argument includes nullable symbols.

Delete the auxiliary $\widetilde{S}$ and use its typed successors $(S_0)_\mu$ as a finite initial set. The type $\mathbf{1}_\sharp$ already accounts for λ when it belongs to L; no extra initial symbol is necessary. A productionless initial symbol handles the empty language. Taking the union over all finite typings gives the inclusion.

For properness over $\{a,b\}$, use the Dyck example also discussed by Clark [25]. Every $x \in D_1$ satisfies $uxv \in D_1$ iff $uv \in D_1$, so $x \equiv_{D_1} \lambda$. Hence $S \to aSbS \mid \lambda$ is congruential, while Corollary 9.5 gives $D_1 \notin \mathsf{RS}(\{a,b\})$. □

10 Conclusion

Qualitative learnability is insensitive to grammatical normal forms, whereas quantitative data bounds reveal the algebraic structure of the typing. The exponential separation concerns typed thickness across BUNF presentations and characteristic samples for set-driven operators, not convergence of arbitrary Gold learners. The window–group hypothesis is sufficient, not claimed necessary; characterizing $\mathsf{RS} \cap \mathrm{CFL}$ remains open.

Data availability

No empirical data were used for the research described in this article.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation and revision of this manuscript, the author used OpenAI's ChatGPT and Anthropic's Claude to assist with language editing, manuscript organization, and the examination of mathematical arguments. The author reviewed and edited the AI-assisted content and takes full responsibility for the results and proofs presented in this article.

A Fixed-window witness bounds and thickness-preserving normalization

A.1 Proof of Lemma 7.1

Proof of Lemma 7.1. Put $m = k + \ell$, $q = |\mathsf{H}|$, $N = |V|$, $\bar{\tau} := \tau_G + 1$, $D_{\mathrm{dep}} := (m+2)Nq$ and $W_{\mathrm{ess}} := m + q$. Fix any retained typed copy A_μ, including a copy of the original start symbol. If $\mu = \mathbf{1}_\sharp$, its only yield is λ. Otherwise choose a positive $w_0 \in L_G(A)$ of type μ and put $\eta := (s_{k,\ell}(w_0), g(w_0))$. Select a shortest *nonempty* word $w \in L_G(A)$ in this $(s_{k,\ell} \times g)$-fiber, and a derivation tree for w with the fewest nonterminal nodes. The kernel hypothesis shows $\widehat{s}(w) = \mu$, and $|\omega(A_\mu)| \le |w|$. If $|w| < m$, the claimed bound is immediate.

Since $|w| \ge m = k+\ell$, the first k and last ℓ positions are disjoint (either window block may be empty). Protect these letters and one additional letter as a *nonemptiness anchor*; it may already lie in a window block. Let Prot be the protected position set, with $1 \le d := |\mathrm{Prot}| \le m + 1$. In particular, the anchor also prevents a shortening to λ when $m = 0$. Any replacement retaining the protected letters in order and the group value preserves nonemptiness and the prefix–suffix window, hence the $(s_{k,\ell} \times g)$-value.

Consider all nonterminal occurrences of the chosen BUNF derivation, including unit and nullable nodes. At each node z let $\mathrm{nt}(z)$ be its nonterminal label, $y(z)$ its possibly empty

yield and Mark(z) the protected positions contained in its span. Marks are nested along each ancestor chain and assume at most $d+1$ different values. The triple $(\mathrm{Mark}(z), \mathrm{nt}(z), g(y(z)))$ cannot recur along a chain. Indeed, equal nonterminal labels permit grafting the lower derivation at the higher occurrence. Equality of $g(y(z))$ preserves the group value of the entire word by the homomorphism property, and equality of marks retains all protected letters in order. The new word lies in the same nonempty fiber, and its length cannot increase. The grafted tree has strictly fewer nonterminal nodes. Strict shortening contradicts minimal word length, and equal length contradicts the tie break. Thus the tree has depth at most $(d+1)Nq \le D_{\mathrm{dep}}$.

Call a node *essential* when its yield span contains a protected position or its yield has length exceeding $\bar{\tau}$. At any depth at most d nodes meet protected positions. If there were $d+q$ essential nodes at the same depth, choose q disjoint unprotected long-yield nodes, ordered left-to-right, with yields $a_1, \dots, a_q$ longer than $\bar{\tau}$. By the definition of ordinary thickness choose b_i in the terminal language of each corresponding nonterminal, $|b_i| \le \tau_G$; the b_i may be λ.

Write $w = c_0 a_1 c_1 \cdots a_q c_q$. Let

$$r_j = g(a_1 c_1 \cdots a_j c_j), \quad t_j = g(b_1 c_1 \cdots b_j c_j), \quad x_j = t_j r_j^{-1}, \qquad x_0 = 1_{\mathsf{H}}.$$

Among the $q+1$ values $x_0, \dots, x_q$ two agree, say $x_i = x_j$ for $i < j$. Writing $r_j = r_i r_{i:j}$ and $t_j = t_i t_{i:j}$ for the corresponding ordered interval products, cancellation in $t_i r_i^{-1} = t_i t_{i:j} r_{i:j}^{-1} r_i^{-1}$ gives $t_{i:j} = r_{i:j}$, that is,

$$g(b_{i+1} c_{i+1} \cdots b_j c_j) = g(a_{i+1} c_{i+1} \cdots a_j c_j).$$

Replacing the indicated disjoint subtrees preserves the ordered group value, the protected letters, and nonemptiness, while strictly shortening the word. The argument does not assume H abelian. Thus every depth has at most $W_{\mathrm{ess}} = m + q$ essential nodes.

Essential nodes are ancestor-closed and there are at most $W_{\mathrm{ess}} D_{\mathrm{dep}}$ of them. Since BUNF has at most two children per node, the number of maximal nonessential subtrees is at most $1 + 2W_{\mathrm{ess}} D_{\mathrm{dep}}$. Each contributes at most $\bar{\tau}$ letters. All remaining terminal positions are protected (at most d of them), so

$$|w| \le d + (1 + 2W_{\mathrm{ess}} D_{\mathrm{dep}})\bar{\tau} \le m + 1 + [1 + 2(m+q)(m+2)Nq](\tau_G + 1).$$

The same argument treats positive start copies; empty-type copies yield only λ. This proves the lemma. □

A.2 Thickness-preserving BUNF normalization

Proof of Proposition 7.3. Let G_* be reduced and put $n := |G_*|$, $\bar{\tau} := \tau_{G_*} + 1$. First isolate terminals in right-hand sides of length at least two, using wrappers $W_a \to a$, and then binarize. Retain all nullable and unit rules. These standard operations preserve all original nonterminal languages [22, Section 3.1] [23, Sections 7.1 and 7.4.2]. Each auxiliary binarization symbol is fresh for its original production and represents a contiguous block of that production's right-hand side, containing at most $O(n)$ original symbols. Substituting shortest yields gives a terminal word of length $O(n\bar{\tau})$. The original nonterminal languages and their shortest yields are unchanged. Trim unreachable or unproductive symbols. The resulting BUNF grammar is equivalent, has polynomial size and $\tau_G = O(n(\tau_{G_*} + 1))$. □

B Proof of the linear-spine BUNF normalization

Proof of Proposition 8.1. A linear CFG has rules $A \to uBv$ or $A \to w$, for terminal words $u, v, w \in \Sigma^*$. Retain $A \to B$ and $A \to \lambda$. No nullable or unit elimination is needed.

For each $a \in \Sigma$ introduce a fresh terminal wrapper $W_a \to a$, and choose all other auxiliary spine symbols freshly per production. For a rule $A \to a_1 \cdots a_m B b_1 \cdots b_n$ with $m + n > 0$, form the chains

$$C_0^{\mathrm{R}} := B, \quad C_j^{\mathrm{R}} \to C_{j-1}^{\mathrm{R}} W_{b_j} \qquad (1 \le j \le n),$$
$$C_0^{\mathrm{L}} := C_n^{\mathrm{R}}, \quad C_i^{\mathrm{L}} \to W_{a_{m-i+1}} C_{i-1}^{\mathrm{L}} \quad (1 \le i \le m).$$

Identify $C_n^{\mathrm{R}} = A$ when $m = 0$, and $C_m^{\mathrm{L}} = A$ otherwise. For terminal-only $A \to a_1 \cdots a_k$, keep $k = 0, 1$ unchanged. If $k \ge 2$, let $C_1^{\mathrm{T}} = A$ and use fresh $C_2^{\mathrm{T}}, \ldots, C_k^{\mathrm{T}}$ with

$$C_i^{\mathrm{T}} \to W_{a_i} C_{i+1}^{\mathrm{T}} \ (1 \le i < k), \qquad C_k^{\mathrm{T}} \to a_k.$$

Every new binary rule has one wrapper child producing exactly one terminal, and one continuing spine child. The original languages are preserved. Trim to obtain a reduced BUNF grammar of polynomial encoded size. □

C Substitutability of the bracketed doubling family

Set $L_0 = \{\mathtt{a}\}$. The languages $L_n = L(G_n)$ from Proposition 6.4 satisfy $L_n = \lceil L_{n-1} \rceil \cup \lfloor L_{n-1} L_{n-1} \rfloor$ for $n \ge 1$, and $L_{G_n}(E_i) = L_i$ for $0 \le i \le n$. Induction on i shows that every word of L_i starts with exactly i opening brackets followed by $\mathtt{a}$. We prove substitutability directly by a deterministic, reversible top-down parser.

Parser and stack properties. Fix $n \ge 1$. A stack is a word over $E_0, \ldots, E_n$ and the closing terminals $\rceil$ and $\rfloor$, with its top on the left. The empty stack is λ. For an arbitrary stack ψ and $1 \le i \le n$, all transitions are

$$E_0\psi \xrightarrow{\mathtt{a}} \psi, \quad E_i\psi \xrightarrow{\lceil} E_{i-1}\rceil\psi, \quad E_i\psi \xrightarrow{\lfloor} E_{i-1}E_{i-1}\rfloor\psi,$$
$$\rceil\psi \xrightarrow{\rceil} \psi, \qquad \rfloor\psi \xrightarrow{\rfloor} \psi.$$

Write $\sigma \xrightarrow{w} \theta$ for the run reading w from σ to θ. These transitions are deterministic. For each fixed input letter they are also injective: its output either reveals the unique expanded E_i and the untouched suffix, or results from deleting a top symbol determined by that input letter. Hence every word acts as a partial injection: if $\sigma \xrightarrow{v} \theta_{\mathrm{out}}$ and $\theta \xrightarrow{v} \theta_{\mathrm{out}}$, then $\sigma = \theta$.

Assign weight 4^i to E_i and weight 1 to either closing marker, and sum over stacks. Every transition strictly decreases weight: the $\mathtt{a}$-transition and both closing-marker pops remove weight 1, while $4^i > 2 \cdot 4^{i-1} + 1 > 4^{i-1} + 1$ for $i \ge 1$. In particular $\sigma \xrightarrow{w} \sigma$ implies $w = \lambda$. Two elementary stack facts will be used. First, in a run from $\upsilon\vartheta$ with $\upsilon \ne \lambda$, until its first visit to ϑ the stack has the form $\upsilon'\vartheta$ with $\upsilon' \ne \lambda$ and its moves affect only υ'. If it visits ϑ, the run splits as $\upsilon \xrightarrow{w_1} \lambda$ followed by $\vartheta \xrightarrow{w_2} \theta$, where $w = w_1 w_2$; otherwise it ends at $\upsilon'\vartheta$ with $\upsilon' \ne \lambda$. The visit to ϑ can occur at most once by strict weight decrease. Second, since only the top symbol is rewritten, a nonempty stack whose last symbol is $\rceil$ (respectively $\rfloor$) retains that last symbol until the entire stack is popped.

Lemma C.1. *For $0 \le i \le n$, every stack ψ and $w \in \Gamma_{\mathrm{br}}^*$, $E_i\psi \xrightarrow{w} \psi$ if and only if $w \in L_i$. Moreover, if $\sigma \xrightarrow{w} \theta$ and $w \in L_i$, then $\sigma = E_i\psi$ and $\theta = \psi$ for some stack ψ.*

Proof. If $E_i\psi \xrightarrow{w} \psi$, the suffix property forces a first visit to ψ (otherwise the final stack would be $\upsilon'\psi$ with $\upsilon' \ne \lambda$). The remaining input must be empty by strict weight decrease. Thus this run is exactly a run $E_i \xrightarrow{w} \lambda$ with the suffix ψ retained. Conversely, any such run from E_i lifts to one from $E_i\psi$.

We prove $E_i \xrightarrow{w} \lambda$ iff $w \in L_i$ by induction on i. For $i = 0$, the only possible run is the single $\mathtt{a}$-transition. For $i \ge 1$, a run must start with $\lceil$ or $\lfloor$. In the unary case the stack becomes $E_{i-1}\rceil$.

The suffix property forces the child E_{i-1} to be completely processed before the final $\rceil$ is popped. Hence $w = \lceil z \rceil$ with $E_{i-1} \xrightarrow{z} \lambda$, equivalently $z \in L_{i-1}$ by induction. In the binary case the stack becomes $E_{i-1}E_{i-1}\rfloor$. Applying the same suffix property successively to its first and second E_{i-1} forces $w = \lfloor z_1 z_2 \rfloor$ with $z_1, z_2 \in L_{i-1}$. Both decompositions conversely give accepting runs, proving the equivalence.

For the final assertion, the first i letters of w are openers, followed by $\mathtt{a}$. The first opener (or $\mathtt{a}$ if $i = 0$) requires $\sigma = E_k\psi$. Each opener decreases the exposed index by one, so the following $\mathtt{a}$ requires $k = i$. The established equivalence and determinism give $\theta = \psi$. □

Lemma C.2. *If $0 \le i \le n$ and $w, w' \in L_i$, then $w \equiv_{L_n} w'$.*

Proof. In any run $E_n \xrightarrow{p} \sigma \xrightarrow{w} \theta \xrightarrow{q} \lambda$, Lemma C.1 forces $\sigma = E_i\psi$ and $\theta = \psi$. The same lemma permits $E_i\psi \xrightarrow{w'} \psi$. Thus $(p, q) \in \mathcal{D}_{L_n}(w)$ implies $(p, q) \in \mathcal{D}_{L_n}(w')$; symmetry gives equality. □

Lemma C.3 (branch synchronization)**.** *If $\sigma \xrightarrow{x} \theta$ and $\sigma \xrightarrow{y} \theta$ for nonempty words with different first letters, then $\sigma = E_i\psi$ for some $1 \le i \le n$ and some stack ψ, and there are decompositions $x = wx'$ and $y = w'y'$ with $w, w' \in L_i$ such that $\psi \xrightarrow{x'} \theta$ and $\psi \xrightarrow{y'} \theta$.*

Proof. Only $E_i\psi$ admits two initial moves, reading $\lceil$ and $\lfloor$. Say these occur for x and y, leaving $E_{i-1}\rceil\psi$ and $E_{i-1}E_{i-1}\rfloor\psi$. Until the first visit to ψ, the suffix property and the last-marker property force the respective stacks to have the forms $A\rceil\psi$ and $B\rfloor\psi$. Each then has weight strictly greater than that of ψ. At the first visit to ψ the weight equals that of ψ, and after any further move it is smaller. Thus, if both runs end at the same θ, either both visit ψ or neither does. In the latter case $A\rceil\psi = B\rfloor\psi$ at θ; right cancellation would give $A\rceil = B\rfloor$, impossible because the last markers differ. Both runs therefore visit ψ. Their prefixes up to these visits are in L_i by Lemma C.1, and their remaining suffixes take ψ to θ, as required. □

Proposition C.4. *For arbitrary stacks σ, θ and words $x, y \in \Gamma_{\mathrm{br}}^*$, if $\sigma \xrightarrow{x} \theta$ and $\sigma \xrightarrow{y} \theta$, then $x \equiv_{L_n} y$.*

Proof. Induct on $|x| + |y|$. If $x = \lambda$, then $\sigma = \theta$, so $y = \lambda$ by strict weight decrease; the symmetric case is identical. If $x = cx'$ and $y = cy'$ share their first letter, determinism gives a common state after c, and induction gives $x' \equiv_{L_n} y'$, hence $x \equiv_{L_n} y$. Otherwise Lemma C.3 gives $x = wx'$ and $y = w'y'$ with $w, w' \in L_i$ and $\psi \xrightarrow{x'} \theta$, $\psi \xrightarrow{y'} \theta$. The removed prefixes are nonempty, so induction applies to x', y'. Lemma C.2 gives $w \equiv_{L_n} w'$. The monoid congruence property then gives $wx' \equiv_{L_n} w'y'$, as desired. □

Proof of Proposition 6.4(i). Let $x, y \in \Gamma_{\mathrm{br}}^+$ and $(u, v) \in \mathcal{D}_{L_n}(x) \cap \mathcal{D}_{L_n}(y)$. The two accepting runs from E_n to λ reach the same stack σ after u by determinism. Their stacks θ_x, θ_y before v satisfy $\theta_x \xrightarrow{v} \lambda$ and $\theta_y \xrightarrow{v} \lambda$. Partial injectivity of the word action gives $\theta_x = \theta_y =: \theta$. Thus $\sigma \xrightarrow{x} \theta$ and $\sigma \xrightarrow{y} \theta$. Proposition C.4 gives $\mathcal{D}_{L_n}(x) = \mathcal{D}_{L_n}(y)$. Since L_n is finite and hence context-free, it is Clark–Eyraud substitutable and belongs to $\mathcal{C}^{\mathrm{cf}}_{s_0} \cap \mathcal{C}^{\mathrm{cf}}_{s_{\mathrm{U}}}$. □

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